

NOVUS ORDO

Operator's Manual

The professional crypto trading terminal — how to use every part of it

MARKETS · QUANT DESK · STRATEGY LAB · CRYPTO SWARM · WALLETS

CONSTRUCTED BY

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About this manual

This manual explains how to use every part of Novus Ordo, the professional crypto trading terminal: the market terminal, the Quant desk, the Strategy Lab and its bot builder, the crypto swarm, and the wallet and exchange connections that link it to your accounts. It is written in plain English. Where a technical term is unavoidable, it is explained where it first appears and again in the glossary at the back.

Each chapter starts with what a part of the terminal is for, then shows how to use it step by step. Function codes — short names you type on the command line, such as `VAR` or `BOT` — appear in a box like this so you can find them on screen.

How to read the screenshots

The screenshots in this manual were taken from the real application running on its built-in market simulator and on sample paper-trading data. Prices, balances and results shown in them are illustrations, not real performance.

Designed and constructed by

Quincy Gininda

Novus Ordo — its concept, design, trading systems and this manual — is a construction by Quincy Gininda.

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Exchange, wallet and service names mentioned in this manual (Hyperliquid, Binance, Bybit, OKX, Coinbase, Kraken, Bitget, KuCoin, Luno, Deribit, Phantom, MetaMask, Rabby, Telegram, Yahoo Finance) belong to their owners and are used only to describe how Novus Ordo connects to them. Novus Ordo is independent of them.

Important risk notice

Trading cryptocurrencies, and especially leveraged perpetual futures, carries a high risk of loss, including the loss of all money deposited. Novus Ordo is software: it is not financial advice, an investment adviser or a broker, and it does not hold your money. Backtests and forecasts describe the past and estimate probabilities; they do not guarantee future results. Use paper trading first, start small, and only trade money you can afford to lose. See chapter 11 for the full notice.

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CHAPTER 1

Welcome to Novus Ordo

One terminal for the serious crypto trader: read the market, model it, build and test strategies, and let an autonomous swarm trade them under strict risk control.

What Novus Ordo is

Novus Ordo is a desktop trading terminal for Windows. It connects directly to the major crypto exchanges, shows their order flow in real time, gives you institutional quantitative models, lets you build and backtest trading bots without writing code, and runs an autonomous team of trading agents — the crypto swarm — on your own exchange account.

Everything runs on your computer. Your exchange keys are encrypted on your machine and never leave it; the terminal talks to the exchanges directly.

The four modes

MODE	OPEN WITH	WHAT IT IS FOR
Markets	`F1`	Reading the market: prices, order flow, footprint charts, liquidity, funding, open interest, news. Chapter 3.
Strategy Lab	`F2`	Building, testing and deploying strategies: the bot builder, research runs on published strategies, the backtest builder. Chapter 5.
Quant desk	`F3`	Institutional risk and portfolio models: value at risk, volatility forecasting, CAPM, the efficient frontier, the Kelly criterion, regimes, delta hedging, a neural network, pairs, seasonality and Monte Carlo — each one usable inside a bot. Chapter 4.
Crypto swarm	`Ctrl` + `2`	The autonomous trading team and your accounts: the Command Center, deployed strategies, the order ticket, wallets and alerts. Chapter 6.

Markets, the Strategy Lab and the Quant desk are the three workspaces of **the Terminal** (`Ctrl` + `1`). The **Crypto swarm** is a separate view. The two meet in one place only: the strategies you deploy from the Strategy Lab into the swarm.



The crypto swarm's Command Center: the swarm core in the centre, the account on the left and right, the gauges along the bottom.

Who it is for

- **Active traders** who want to see what is really happening in the market — who is buying, who is selling, where the liquidity sits — rather than price alone.
- **Systematic traders** who want to test ideas on years of data before risking money, with honest statistics that say when an idea has no edge.
- **Anyone who wants automation with control**: bots that trade only within limits you set, start on paper, and report every decision.

What you need

- A Windows 10 or Windows 11 computer (64-bit) and an internet connection.
- For live trading: an account on a supported exchange — Hyperliquid for the crypto swarm (chapter 7 explains how to fund it), or any of the exchanges in chapter 7 for the order ticket and bots.
- For notifications on your phone (optional): a Telegram account.

CHAPTER 2

Getting started

Install the terminal, unlock it, and find your way around the screen in a few minutes.

Installing

Novus Ordo comes in two forms. Use one of them, not both.

DOWNLOAD	HOW TO USE IT
Zip (recommended) NovusOrdo-...-win-x64.zip	Extract it once to a folder such as `C:\Novus Ordo` (not inside the zip viewer), then run **Novus Ordo.exe**. Pin it to the taskbar. It opens in seconds.
Portable NovusOrdo-...-portable.exe	A single file you can run from anywhere. It unpacks itself on every launch, so it starts more slowly, especially while Windows Defender scans it.

First launch on Windows

If Windows SmartScreen shows "Windows protected your PC", choose **More info → Run anyway**. This appears for new applications that are not yet widely downloaded.

The lock screen and your ledger key

Novus Ordo opens on a black lock screen with the emblem and the prompt **ENTER LEDGER KEY**. While it waits, the terminal loads in the background — the database, the key vault, the exchange connectors and the market feeds — and lists each item in the bottom-right corner as it arrives.

1. Type your 8-digit **ledger key**. The terminal opens as soon as the eighth digit is entered.
2. Until the key is entered, nothing private is available: API keys, trading, bots, wallets and Telegram stay sealed by the desktop process itself.
3. Five wrong attempts lock the input for 30 seconds, doubling after each further miss.
4. To change the key: ⚙ **Settings → Ledger key**. You need the current key; the new one must be 8 digits and not a simple pattern.

🔒 **Lock** on the top bar returns to the lock screen whenever you leave the computer. Market data and running bots keep working underneath, and the lock screen shows how long the terminal has been locked and what is running.

Finding your way around



The terminal: views and workspaces on the top bar, the command line and quote tape, the function keys, then the panels.

- **Top bar** — the two views (▤ **Terminal** and **Crypto swarm**), the three workspaces (**MARKETS**, **STRATEGY LAB**, **QUANT DESK**), the current coin and venue, the live price, 🔌 **Connect**, the theme, ⧉ **Screens**, 🔒 **Lock** and ⚙ **Settings**.
- **Command line** — type a function code and press `Enter`. Put a coin first to switch the market at the same time: `ETH GP` opens the price graph for ETH, `SOL BOOK` the order book for SOL. Suggestions appear as you type; pick one with `↑`/`↓`. Type `HELP` for every function.
- **Function keys** — under the command line, the most-used functions of the current workspace, numbered 1 to 10. Click one to open it.
- **Quote tape** — the most traded markets, scrolling, with their 24-hour change.
- **Panels** — every function opens as a panel. Panels sit in tab groups; drag a tab to rearrange, drag the gaps to resize.
- **Status line** — at the bottom: the feed and its speed, the Strategy Lab's state, UTC and local time.

Keyboard shortcuts

KEYS	ACTION
`Ctrl`+`1` / `Ctrl`+`2`	Switch between the Terminal and the Crypto swarm view
`F1` `F2` `F3`	Markets, Strategy Lab, Quant desk
`Ctrl`+`K` or `` ` ``	Jump to the command line
`↑` `↓` then `Enter`	Choose a suggestion on the command line
`Esc`	Leave the command line; close a drawer or dialog
Double-click a tab bar	Maximise that panel group; again to restore

Layouts, themes and several screens

- ✎ **Layout** unlocks the layout: drag panels by their tabs onto another panel to stack them or onto an edge to split; **×** closes a panel; **+ Add panel** adds one. Layouts save automatically.
- ⇄ on any tab bar moves that tab to another workspace or into its own window.
- ⧉ on any tab bar pops the panel out into its own window. ⧉ **Screens** on the top bar lists every screen and window: send any window to any screen as **Fill**, **Full** (full screen) or **Float**. From any view — the crypto swarm included — **Open any tool in its own window** opens any function on another monitor. ⇤ **Back to the terminal** on a pop-out returns it.
- **Themes**: Terminal Amber (black and amber), Workspace, Eikon Blue, Graphite and Paper, from the top bar. Charts follow the theme.



⧉ Screens: the crypto swarm on one screen, the Quant desk's neural network and the bot builder on another.

Start safely

Everything that trades starts in **paper** mode — real prices, simulated money, fees and slippage counted. Keys stay read-only until you tick **Allow trading**. Run on paper first, then on an exchange testnet, then live with a small amount.

CHAPTER 3

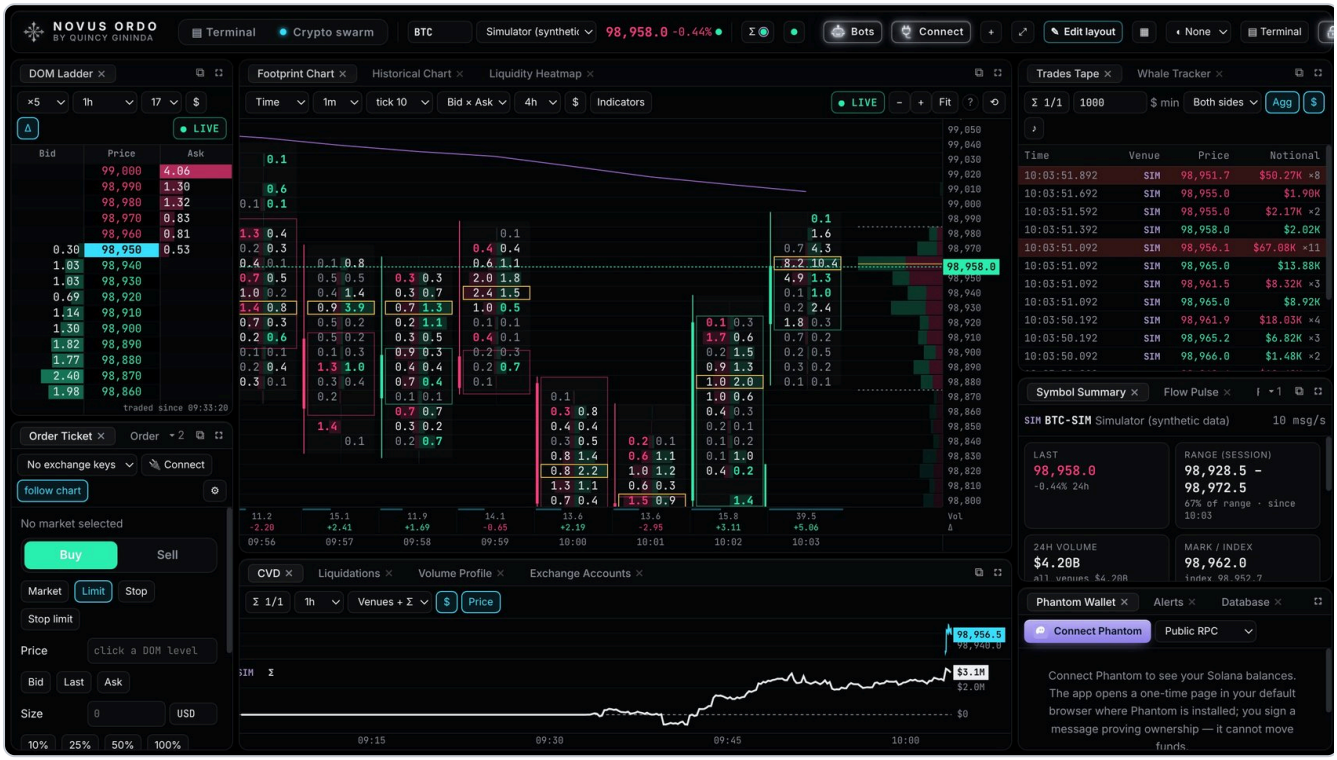
The market terminal

Markets (F1) shows what is happening in the market right now — and who is doing it.

Choosing a market and venues

- **Coin** — type it on the command line (`ETH` then `Enter`) or pick it on the top bar. Every panel follows the coin.
- **Primary venue** — the exchange whose order book and trades the main charts show (for example Binance Futures). Choose it on the top bar.
- **Σ aggregate** — add other venues to see their trades combined with the primary's: the tape, cumulative delta and footprint then show the whole market, not one exchange.

Supported venues: Binance (futures and spot), Bybit, OKX, Hyperliquid, Coinbase, Kraken, Bitget, Bitfinex and Deribit (options), plus any of more than 100 exchanges through CCXT. A built-in **Simulator** produces a realistic synthetic market for learning and testing; it is clearly labelled.



The order-flow layout: the depth ladder (left), the footprint chart (centre), the tape and market summary (right), cumulative delta below.

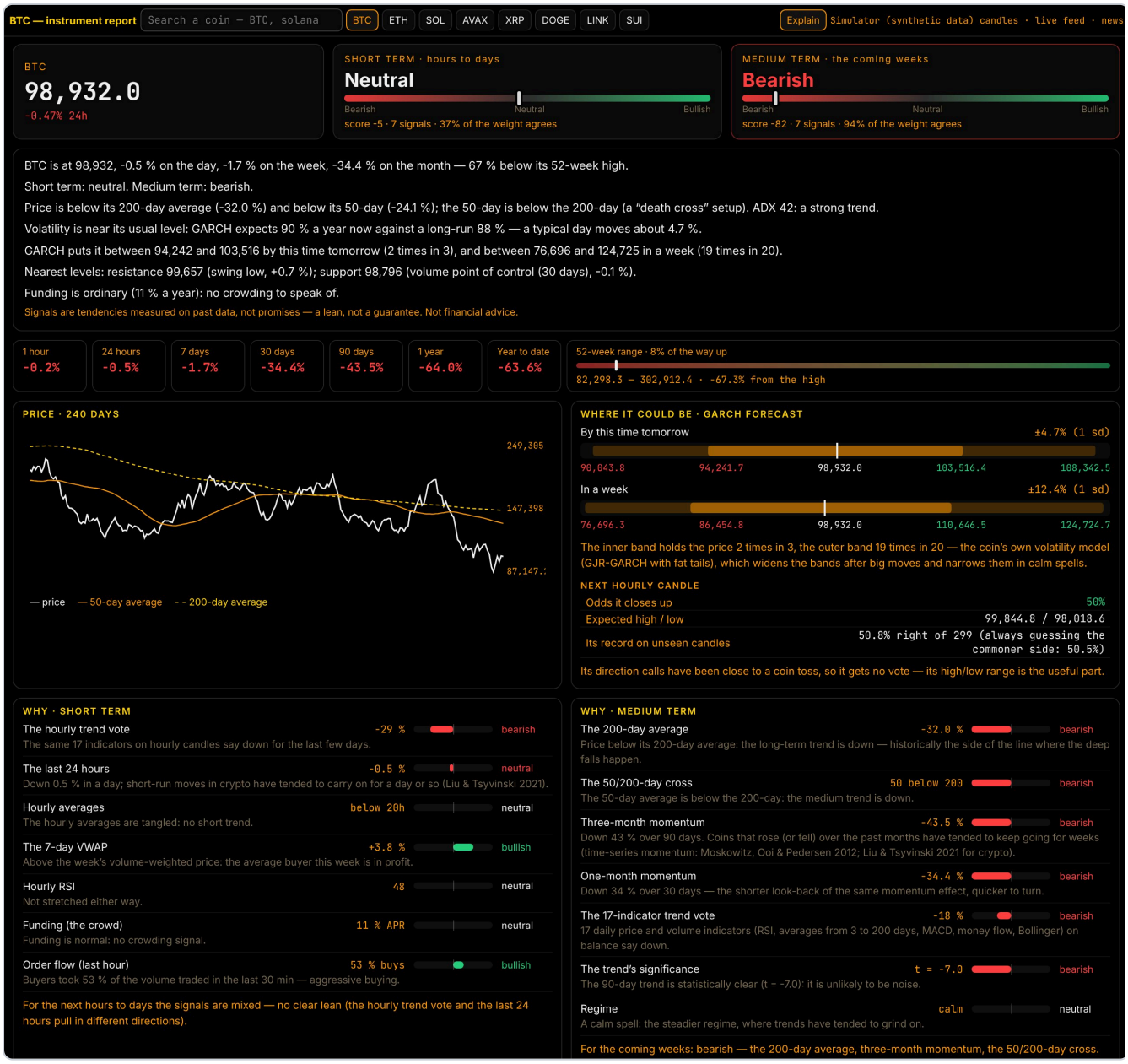
Ask about a coin: the instrument report

Type `ASK SOL` — or simply *ask about solana*, *tell me about bitcoin*, *how is avax doing* — on the command line and press `Enter`. The coin becomes the market and its full report opens in Markets. Coins can be typed by ticker or by name (bitcoin, ethereum, solana, avalanche, chainlink and so on). On the report itself, type any coin in the search box or click one of the quick chips.

The report answers the question **"is it bullish or bearish, and why?"** for two horizons:

- **Short term — hours to days.** The hourly trend vote (17 indicators), the last 24 hours, the hourly averages, the week's VWAP, the hourly RSI, funding (a crowded long is a warning), the last hour's order flow, liquidations, the next-candle model (only if it has beaten chance on candles it never saw) and the tone of the news.
- **Medium term — the coming weeks.** The 200-day average, the 50/200-day cross, three- and one-month momentum, the daily trend vote, the trend's statistical significance, strength against bitcoin and the regime.

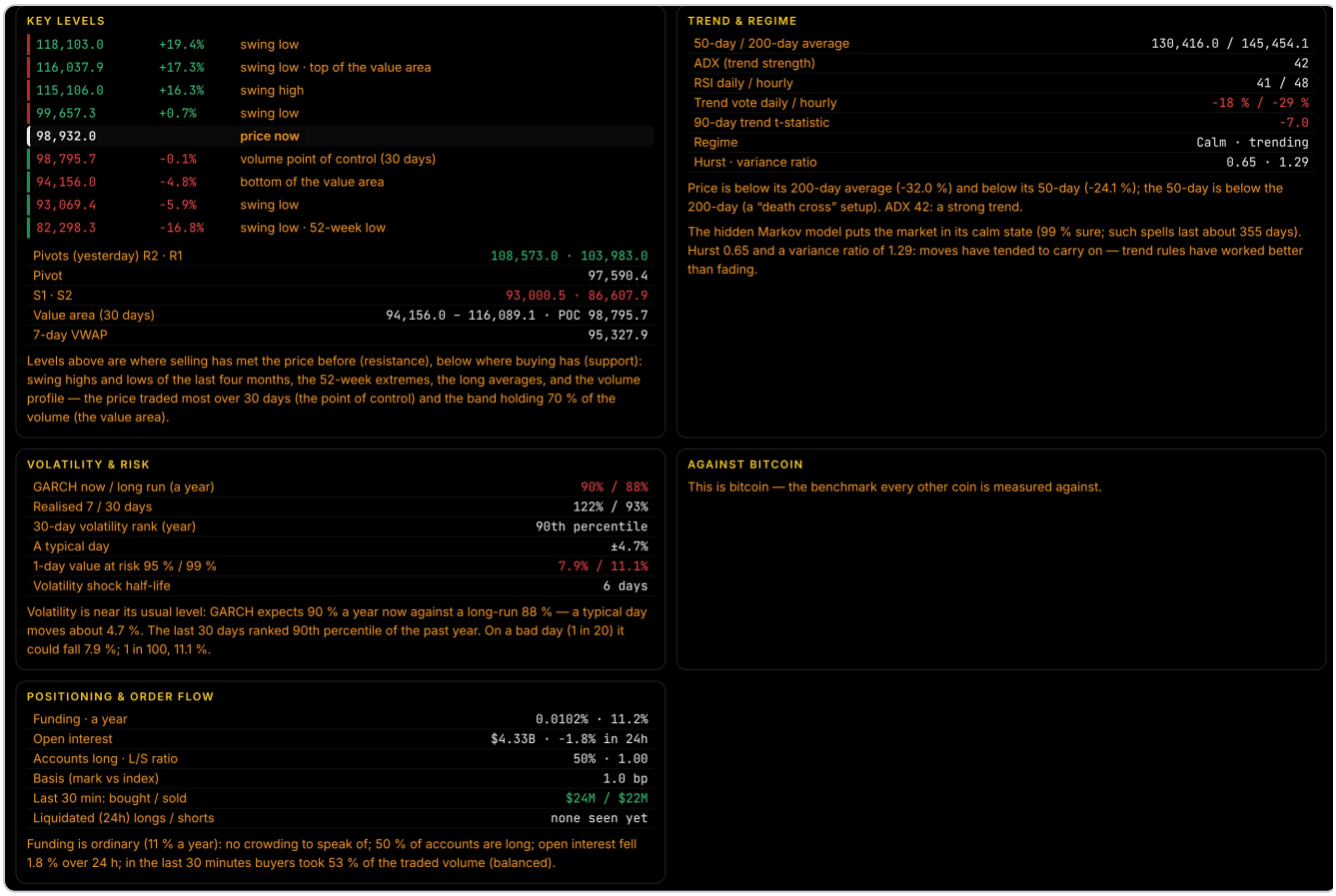
Each signal votes bullish, bearish or neutral with a weight; the balance is the verdict (Bullish, Leaning bullish, Neutral, Leaning bearish, Bearish) shown on a gauge, with how much of the weight agrees. Every vote says why it voted — switch on **Explain** to read it.



ASK BTC (here on the simulator's test data): the two verdicts, the summary in plain words, performance from an hour to a year, the 52-week range, the price with its 50- and 200-day averages, GARCH's ranges for tomorrow and a week, the next hourly candle, and every signal behind each verdict.

Below the verdicts the report sets out everything else worth knowing:

SECTION	WHAT IT TELLS YOU
Performance	The change over 1 hour, 24 hours, 7, 30 and 90 days, a year and the year to date; where the price sits in its 52-week range and how far below the high it is.
Where it could be	GARCH's price ranges for this time tomorrow and in a week: the inner band holds the price 2 times in 3, the outer band 19 times in 20. Plus the next hourly candle's odds, expected high and low, and its record on unseen candles.
Key levels	Resistance above and support below: swing highs and lows of the last four months, the 52-week extremes, the long averages, the volume profile's point of control and value area; yesterday's pivots and the 7-day VWAP.
Trend & regime	The averages, ADX (trend strength), RSI, the trend votes, the trend's t-statistic; calm or turbulent (hidden Markov model) and trending or mean-reverting (Hurst, variance ratio).
Volatility & risk	GARCH volatility now against its long-run level, realised volatility and its rank in the year, a typical day's move, and how much a bad day could cost (value at risk).
Against bitcoin	Beta, correlation and alpha, and whether it has beaten or lagged bitcoin over 30 and 90 days.
Positioning & order flow	Funding and its yearly rate, open interest and its change, the share of accounts long, the basis, the last hour's buying against selling across your venues, and the last day's liquidations.
News	Headlines of the last three days that name the coin, each marked as sounding positive or negative.
In the Strategy Lab	Edge-finder survivors and deployed strategies that trade the coin, with a link to the edge finder.



The rest of the report: key levels either side of the price, trend and regime, volatility and risk, positioning and order flow.

A lean, not a promise.

Every signal in the report is a tendency measured on past data. Use the verdict to frame a decision — then size the trade by its risk, as the Quant desk shows, and never by the verdict alone.

Price and order-flow functions

CODE	FUNCTION	WHAT IT SHOWS AND WHAT IT IS FOR
ASK	Instrument report	Everything about one coin now: bullish or bearish and why, forecasts, levels, risk, positioning, news (see above).
MON	Monitor	The quote board: last price, change, range, volume, funding and open interest for your coins. Your first look at the day.
HMAP	Heat map	The whole market as tiles — size by volume, colour by change. Spot where money is moving at a glance.
GP	Graph price	Candles or market profile (TPO) with VWAP and moving averages, any timeframe. The main price chart.
GIP	Footprint	Each candle broken into prices with the volume bought and sold at each: bid × ask, delta and imbalances. Shows whether buyers or sellers are in control inside a candle.
LHM	Liquidity heat map	Resting orders in the book over time, with trades on top. Find where large orders wait and whether price respects them.
BOOK	Order book	The composite order book across venues.

DOM	Depth ladder	The price ladder with resting size, traded volume and delta at each price. Click a level to send its price to the order ticket.
DPTH	Depth chart	Cumulative bid and ask depth: how much it takes to move price.
TAS	Time & sales	Every trade across venues, sized and coloured; filter by size to see only the large ones.
CVD	Cumulative delta	Buy minus sell volume over time, per venue. Divergence between price and delta warns of weakness.
FLOW	Flow pulse	Buy against sell notional over 30 seconds to 15 minutes.
BLCK	Block trades	Large prints and net whale flow.
LIQ	Liquidations	Forced closures of leveraged positions, long against short. Cascades often mark turning points.
VP	Volume profile	Volume at each price and the value area (where 70 % of trading happened).

Derivatives, volatility and market context

CODE	FUNCTION	WHAT IT SHOWS AND WHAT IT IS FOR
OI	Open interest	Open positions per venue. Rising price with rising OI means new money; with falling OI, positions closing.
FUND	Funding	Funding rates, their yearly rate and the basis. High positive funding means longs are crowded.
OMON	Options	Deribit option trades, implied volatility and the put/call balance.
SPRD	Venue spread	Each venue's price against the primary.
CORR	Correlation	Correlation matrix of the coins, beta to bitcoin and volatility.
HVOL	Volatility	Realised volatility over a day, week and month, the term structure and its rank.
GV	GARCH volatility	The volatility forecast for the hours ahead as a heat calendar (see chapter 4).
FCST	Candle forecast	The next candle's expected range and its odds of closing up, with its track record on unseen data.
DES	Description	The market summary: price, 24-hour statistics, funding, open interest, long/short ratio, basis.
SCRN	Screener	Every perpetual on a venue, sortable.
WL	Watchlist	Your symbols on any venue.
TOP	Top news	Real-time crypto headlines by coin.
WEI	World markets	Stocks, indices, FX and futures from Yahoo Finance.

How to read the order flow

1. **Start wide.** `MON` and `HMAP`: which coins are moving, on what volume.
2. **Find the levels.** `VP` for the value area and the point of control (the busiest price); `LHM` for large resting orders.
3. **Watch the aggression.** `GIP` footprint and `CVD`: a breakout with buyers lifting the offer and delta rising is real; a new high on falling delta is suspect.
4. **Check positioning.** `FUND`, `OI` and `LIQ`: crowded funding and stacked liquidations show where a squeeze could come from.
5. **Act or alert.** Send a price from the `DOM` to the order ticket, or set an alert with `ALRT`.

History and the local database

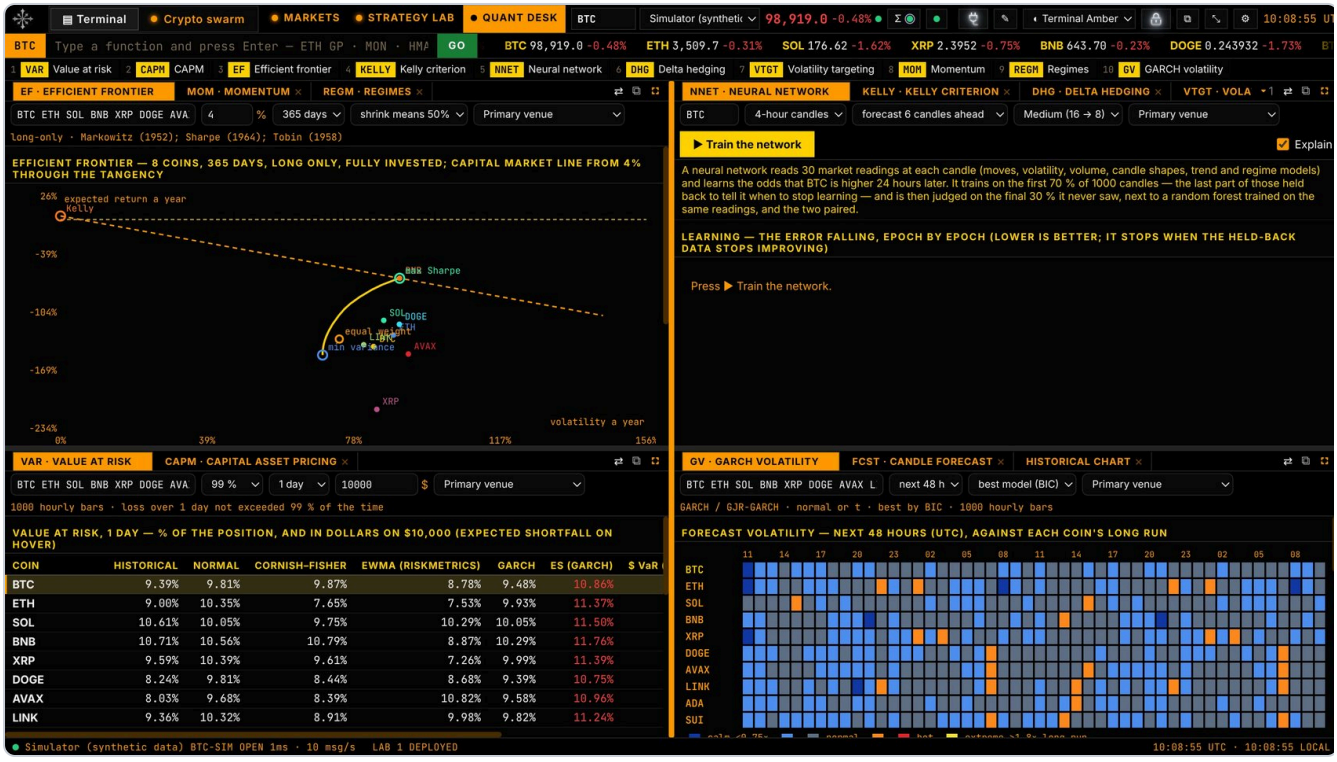
The terminal records the live trades of the markets you watch into a local database, so charts open with history already filled in; any gap since you last ran it is filled from the exchange. Old trades are cleaned out automatically after the retention period (3 days by default). The `DB` panel shows the database size, each recorded market and the retention, and exports a market's trades to CSV.

CHAPTER 4

The Quant desk

Quant desk (F3): the models institutions use to measure risk, size positions and build portfolios — each explained, each one keystroke away.

Every Quant desk panel follows the coin on the board, and the tables cover a list of coins you choose (type them in the coins box, separated by spaces). Each capability below is described by **what it is**, **what it is used for** and **how to use it**.



COIN	HISTORICAL	NORMAL	CORNISH–FISHER	EWMA (RISKMETRICS)	GARCH	ES (GARCH)
BTC	9.39%	9.81%	9.87%	8.78%	9.48%	10.86%
ETH	9.00%	10.35%	7.65%	7.53%	9.93%	11.37%
SOL	10.61%	10.05%	9.75%	10.29%	10.05%	11.50%
BNB	10.71%	10.56%	10.79%	8.87%	10.29%	11.76%
XRP	9.59%	10.39%	9.61%	7.26%	9.99%	11.39%
DOGE	8.24%	9.81%	8.44%	8.68%	9.39%	10.75%
AVAX	8.03%	9.68%	8.39%	10.82%	9.58%	10.96%
LINK	9.36%	10.32%	8.91%	9.98%	9.82%	11.24%

The Quant desk: efficient frontier, value at risk, GARCH volatility and more, side by side.

Risk: value at risk and volatility

VAR Value at risk and expected shortfall

WHAT IT IS The loss that should not be exceeded 95, 99 or 99.5 % of the time over an hour, a day or a week — as a share of the position and in dollars — by five methods: historical, normal, Cornish–Fisher (allowing for skew and fat tails), EWMA (RiskMetrics) and GARCH. Expected shortfall shows the average loss on the days that are worse.

USED FOR Knowing how much a position can realistically lose on a bad day before you take it, and comparing coins by risk. A portfolio line shows what spreading across coins saves.

HOW TO USE Enter your coins, choose the confidence (99 % is standard) and horizon, and your position size in dollars. Check the backtest table: each method's past forecasts against what happened — Kupiec says whether it was breached the right number of times, Christoffersen whether breaches cluster. Trust the methods that pass.

GV GARCH and GJR-GARCH volatility

WHAT IT IS	A volatility forecast: GARCH models how volatility clusters (calm follows calm, storms follow storms); GJR-GARCH adds the extra jump in volatility after falls. Shown as a heat calendar of the hours ahead against each coin's long-run level, with the news-impact curve.
USED FOR	Seeing a storm coming before it arrives: widen stops, cut size, or wait when the forecast is hot; trade breakouts when volatility is expanding.
HOW TO USE	Choose the model (best by BIC, GARCH or GJR-GARCH) and the horizon. Blue hours are calmer than normal, orange and red hotter.

VTGT Volatility targeting and position sizing

WHAT IT IS	Volatility measured seven ways (close-to-close, Parkinson, Garman–Klass, Rogers–Satchell, Yang–Zhang, EWMA, GARCH) and the position size that runs at a target volatility on your account.
USED FOR	Keeping risk steady: a smaller position when the market is wild, a bigger one when it is calm. A chart compares buy-and-hold with the same coin volatility-targeted.
HOW TO USE	Set your target (for example 40 % a year) and account size; read the position for each estimator. A Kelly calculator is included.

DHG Delta hedging and option greeks

WHAT IT IS	Black–Scholes price and greeks for a call, put or straddle on the coin, priced at the GARCH forecast, realised volatility or your own; the hedge to hold on the perpetual now; a strike ladder; and a backtest of selling or buying the option daily and hedging it every 1, 4, 12 or 24 hours.
USED FOR	Understanding options risk and the volatility premium, and knowing exactly how much of the coin to hold against an option position.
HOW TO USE	Choose the option (strike, days, size, bought or sold) and the volatility to price it with; read the delta and the hedge; compare hedging frequencies in the backtest.

Portfolio: CAPM, efficient frontier and Kelly

CAPM Capital asset pricing model

WHAT IT IS	Each coin measured against the market (bitcoin or an equal-weight index): beta (how much it moves with the market), Jensen's alpha and its t-statistic (return beyond what its beta explains), R^2, the return CAPM expects for that beta, Treynor and information ratios, Sharpe and the security market line.
USED FOR	Telling a coin's own performance from simply riding the market, and finding coins that add something beyond bitcoin.
HOW TO USE	Choose the market and the period. A coin above the security market line did better than its risk implied; a high R^2 means it mostly follows the market.

EF Efficient frontier (Markowitz)

WHAT IT IS	The long-only mixes of your coins with the best expected return for each level of risk, with the minimum-variance portfolio, the maximum-Sharpe (tangency) portfolio, the capital market line, equal weight and the Kelly portfolio — and their weights.
USED FOR	Deciding how to split money across coins rather than guessing.
HOW TO USE	Enter the coins and the risk-free rate; choose how much to shrink expected returns toward the average (past returns are noisy, so 50 % is a sensible start). Read the weights in the table.

KELLY Kelly criterion

WHAT IT IS	The leverage that grows capital fastest in the long run for each coin, and half of it; the growth curve by leverage (it peaks at full Kelly and falls back to nothing at twice it); the growth-optimal portfolio under a leverage cap; and Kelly on the crypto swarm's own trade record.
USED FOR	Sizing bets so that you grow without risking ruin. Most professionals use half Kelly or less.
HOW TO USE	Read full and half Kelly per coin; for a strategy, enter its win rate and average win in R to get its Kelly bet.

Market behaviour: momentum and regimes

MOM Momentum

WHAT IT IS	Returns from a week to six months, the month's move in its own standard deviations, the 60-day trend t-statistic, the Kalman-filter slope, a composite signal and each coin's rank against the others.
USED FOR	Finding the coins with the strongest and weakest trends — the basis of trend-following strategies.
HOW TO USE	Sort by the composite signal; a high trend t-statistic means the trend is clear, not noise.

REGM Regimes

WHAT IT IS	Whether a coin is trending, mean-reverting or random (Hurst exponent, variance ratio, Ornstein–Uhlenbeck half-life and s-score), and whether it is in a calm or turbulent state (a two-state hidden Markov model with the odds hour by hour).
USED FOR	Matching the strategy to the market: trend strategies in trending regimes, dip-buying in mean-reverting ones, smaller size in turbulence.
HOW TO USE	Read the character column for each coin and the odds of turbulence before choosing a strategy.

FCST Candle forecast

WHAT IT IS A machine-learning forecast of the next candle from candle shapes, volatility and momentum: its expected high and low and the odds it closes up, with its record on data it never saw.

USED FOR A second opinion on the next move — trusted only as far as its track record shows.

HOW TO USE Compare its hit rate on unseen data with simple guessing before giving it weight.

Artificial intelligence: the neural network

NNET Neural network

WHAT IT IS A small neural network (two hidden layers) trained on a coin's candles to forecast whether price will be higher a chosen number of candles later. It reads around 30 market measurements at each candle and stops learning when its latest training data stops improving, so it does not memorise.

USED FOR Testing whether a learned model finds anything in a market — compared fairly with a random forest on the same inputs, and with the two paired.

HOW TO USE Choose the coin, candle size, how far ahead to forecast and the network size; press **Train the network** and watch the error fall. Read the table "on the 30 % it never saw": a model is only useful if it is right more often than guessing. Then press **⇢ Use in a bot** to run this network inside a strategy (see the end of this chapter). The panel also shows what the network looks at most and how well its odds held up.



NNET: the learning curve, the forecast now, and the results on unseen data against a random forest and the two paired.

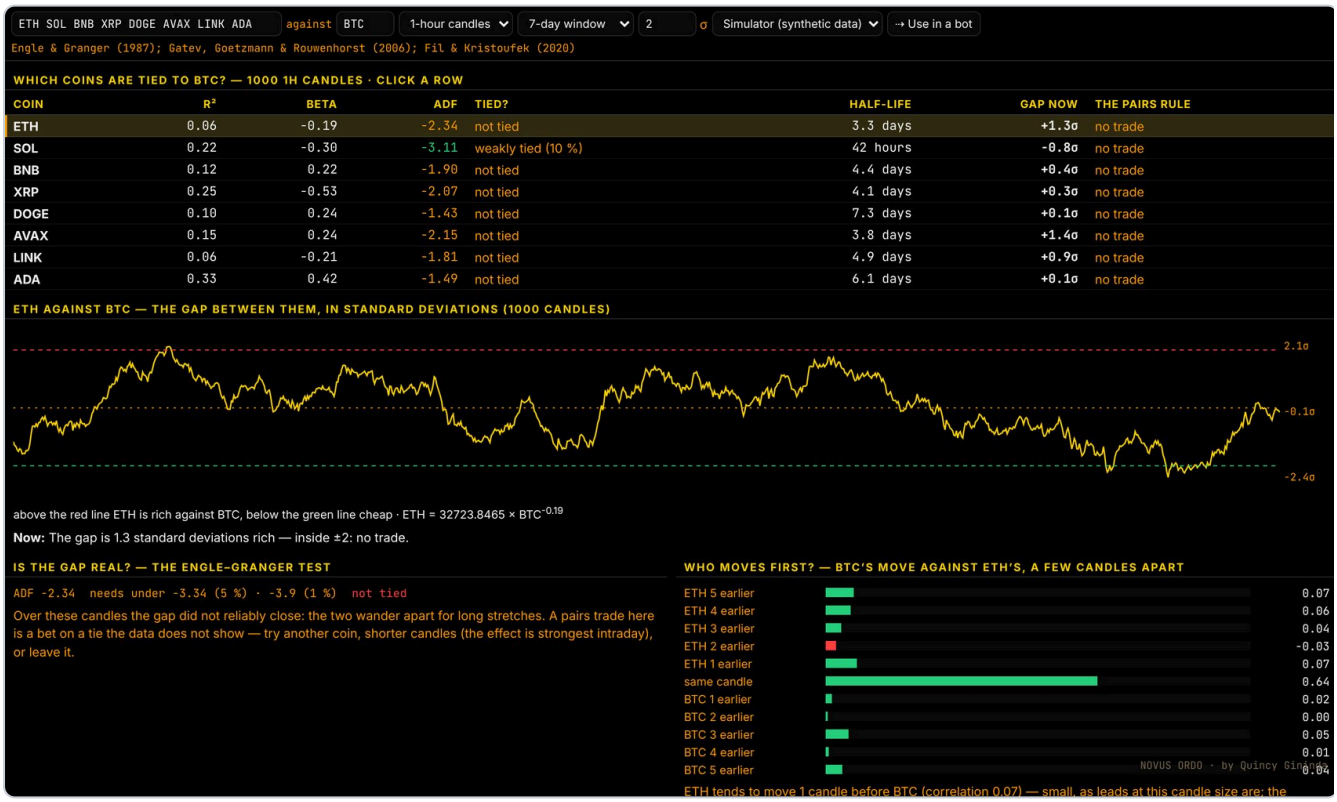
Finding and sizing an edge: pairs, seasons, Monte Carlo

PAIR Pairs and cointegration

WHAT IT IS A screen of which coins are tied to bitcoin (or to any coin you name): how much of each coin's moves bitcoin explains (R^2), the hedge ratio (beta), a statistical test of whether the gap between the two keeps closing (Engle–Granger), how long half of a gap takes to close, where the gap stands now in standard deviations, and who tends to move first.

USED FOR Finding pairs worth trading against each other — buying the coin that is cheap against bitcoin and selling bitcoin, so the market's direction cancels out — and avoiding pairs whose "tie" is not in the data.

HOW TO USE Enter your coins; read the **TIED?** column first ("tied" at the 5 % or 1 % level is what you want) and the half-life (shorter is better). Click a coin for its gap chart. Hourly and shorter candles show the effect best; daily candles rarely do.



PAIR on the simulator, whose coins are independent of each other — so it rightly reports "not tied" and "no trade". On real markets, tied pairs show a strongly negative ADF and a short half-life.

SEAS Seasonality

WHAT IT IS A map of a coin's average return by hour of the day and day of the week (UTC), on a year of hourly candles, with a t-statistic for every hour and weekday.

USED FOR Checking calendar patterns — the published "21:00 to 23:00 UTC" effect in bitcoin, weekend behaviour, your own hunches — before trading them.

HOW TO USE Green cells rose on average, red fell. Read the **T** column: with 24 hours to look at, one will look special by luck, so the panel only says an hour "stands out" past 3.08. Click an hour to select it.



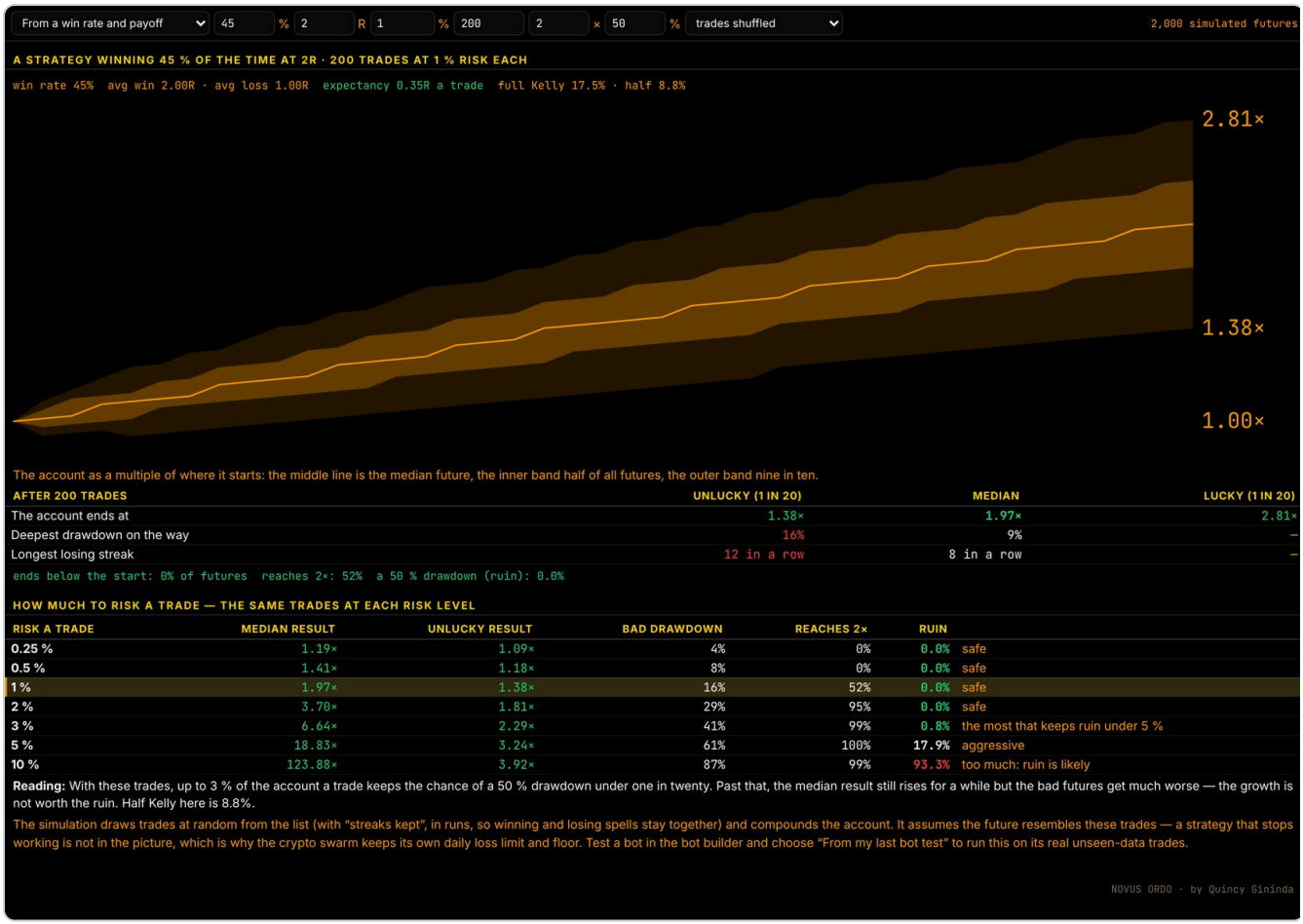
SEAS: the weekday-by-hour map, the hours ranked by how sure their average is, and the plain reading underneath. Here an hour is picked (21:00) to send to the bot builder.

MC Monte Carlo on a strategy's trades

WHAT IT IS A strategy's trades reshuffled into 2,000 possible futures: the range of where the account could end, the deepest drawdown on the way, the longest losing streak, and the odds of reaching a goal or of a ruinous drawdown — worked out again at seven levels of risk per trade.

USED FOR Deciding how much to risk a trade before any money is at stake, and knowing in advance how bad a normal bad run looks, so you do not abandon a sound strategy in one.

HOW TO USE Choose **From my last bot test** to use the real trades of the bot you last tested in the bot builder, or type a win rate and payoff. Set the risk per trade, the number of trades, your goal and what drawdown counts as ruin. Read the ladder: the highest risk that keeps ruin under 5 % is marked.



MC: the fan of possible account paths, the unlucky, median and lucky outcomes, and the same trades at each risk level.

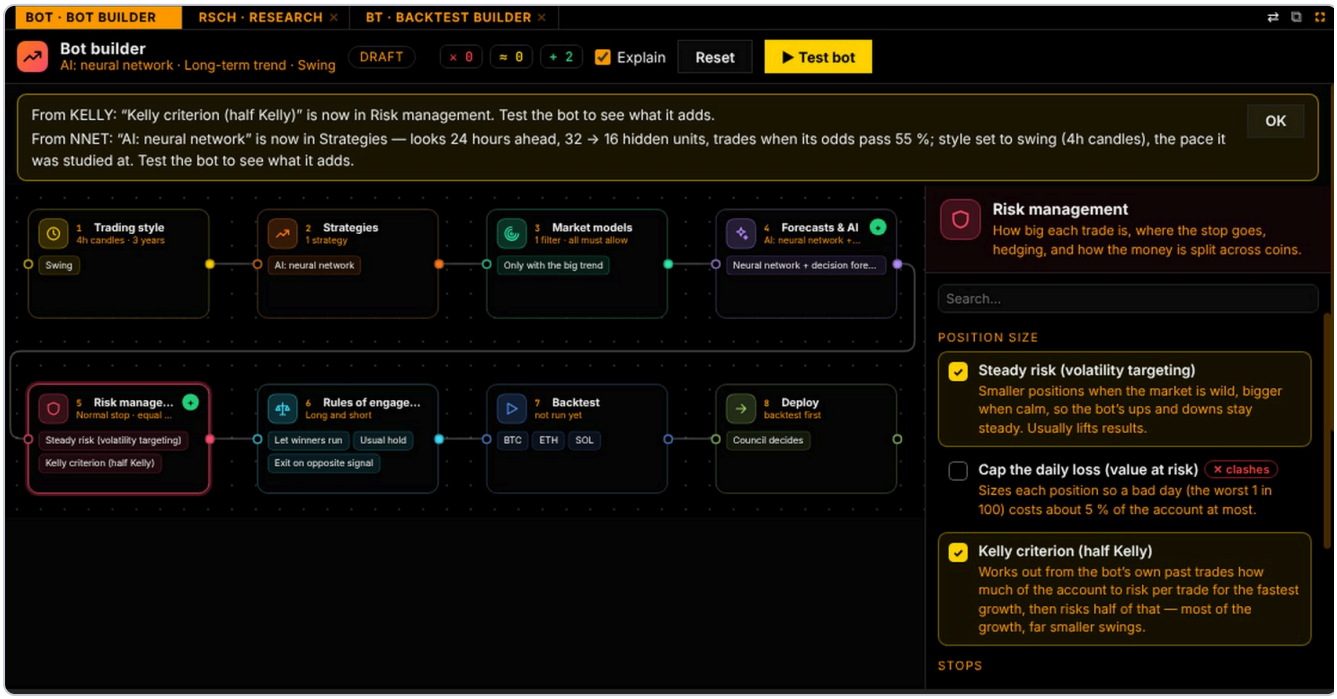
Using a Quant desk model inside a bot

The models on the Quant desk are not separate from your strategies. Every Quant desk panel has a ⇢ **Use in a bot** button: it sends the model on screen — with the settings you are looking at — into the Strategy Lab's bot builder, where it becomes part of the bot you are building.

1. Set the panel up the way you want the model: for example in `NNET`, the coin, the candle size, how far ahead it forecasts and the network size.
2. Press ⇢ **Use in a bot** and choose what it should be in the bot (the choices are listed below).
3. The Strategy Lab opens on the bot builder. A note at the top says what arrived, and the settings appear beside the pick, where you can change them or press **reset**.
4. Press ▶ **Test bot**. The bot is tested like any other, on data it never saw.

FROM	WHAT IT CAN BECOME IN THE BOT
`NNET`	The bot's **strategy** (replacing or joining the others), a **second opinion** that must agree, or the **AI decision maker** (alone, or paired with the decision forest) — with the same network size, horizon, candle size and coin.
`GV`	A filter for calm or stormy volatility, stops that follow the GARCH forecast, or momentum scaled by GARCH volatility.
`VAR VTGT KELLY`	A position-sizing rule: a value-at-risk cap, volatility targeting at the target on the panel, or half Kelly.

DHG CAPM	The bitcoin hedge (market-neutral); CAPM also offers the bitcoin market-trend filter.
EF	How the bot's money is split across coins: efficient frontier, minimum variance or inverse volatility.
MOM REGM FCST	Momentum and Kalman-trend strategies; Hurst, variance-ratio and hidden-Markov filters; the Ornstein–Uhlenbeck strategy; the next-candle forecast as a strategy or second opinion.
PAIR	A hedged pairs trade on the coin, window and threshold on screen; or the "bitcoin leads" strategy.
SEAS	The hour you picked on the map as a strategy; weekday, weekend and trading-hours filters.



The bot builder after two sends from the Quant desk: the note at the top says what arrived, the neural network is the bot's strategy, and Kelly sizing has joined the risk block.

Why the bot retrains the network

When a neural network goes from NNET into a bot, the bot uses the *same network* — the same inputs, size and horizon — but the Strategy Lab trains it again on the training years only. The copy trained on the desk has already seen the recent period, and a test is only honest if the model has never seen the data it is judged on.

CHAPTER 5

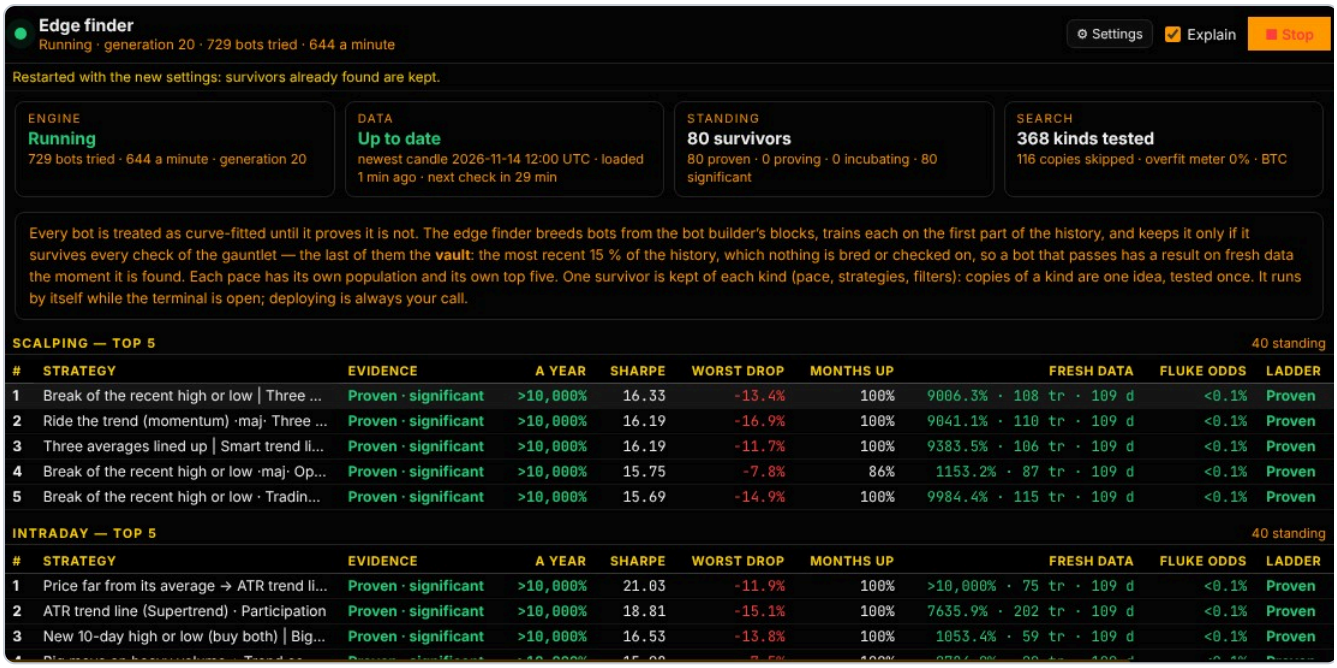
The Strategy Lab

Strategy Lab (F2): build a strategy, test it honestly on years of data, and deploy it to the crypto swarm.

The edge finder: strategies found for you

The edge finder (`FARM`) is a strategy farm that runs by itself for as long as the terminal is open — for days, if you leave it. You do not need to build and test bots one at a time: it builds them from the bot builder's blocks — every strategy and filter, and the Quant desk's models — tests them, and keeps a **top five for each pace**: scalping, intraday and swing. It treats every bot as curve-fitted until the bot proves it is not.

1. **Breed.** Bots are built from the blocks — one to three strategies, joined in any of five ways — and scored on the first part of the history. Each pace has its own population, bred side by side. The best are crossed, *merged* (the strategies of two bots joined into one) and varied into the next generation, and new random bots keep arriving.
2. **The unseen 30 %.** A bot that did well in training is tested on a later stretch of the history, which it never saw. It must make money there, and do at least half as well as in training.
3. **Steady.** Not one lucky stretch, coin or trade: it must be in profit in at least two of the three thirds of the unseen stretch, on at least half of its coins, and still in profit with its best 5 % of trades taken away.
4. **Double costs.** It is tested again with twice the fees and slippage. An edge that only exists at perfect prices is not an edge.
5. **100 random bots.** It must beat at least 75 of 100 bots that make the same number of trades at random.
6. **Its settings nudged.** Every setting is moved one step each way. If fewer than half of those versions still make money, its result was luck in its exact settings, and it is thrown out.
7. **1,000 shuffles.** Its trades are put in 1,000 random orders. If the unlucky orders cause too deep a drawdown, it is thrown out.
8. **The vault.** The most recent 15 % of the history is locked away: no bot is bred or checked on it. A bot that passes everything above is tested on it once. If it loses there, it is out — and so is its kind. This is why a survivor has a result on fresh data the moment it is found, instead of weeks later.
9. **Significant after every kind tried.** The more ideas are tried, the better the luckiest of them looks. So the bar for "significant" rises with every kind of bot that reaches the unseen data: of those called significant, at most about 1 in 10 should be flukes. Each survivor shows its **fluke odds**. Those that miss the bar are **candidates**: still listed, and still tested on fresh data.
10. **Fresh data.** A survivor's record starts with the vault and carries on into the new candles that arrive, and it climbs a ladder: incubating (fewer than 5 trades) → proving (5 to 14, in profit) → proven (15 or more, in profit). One that loses fades and is dropped.



The edge finder running (here on test data). The strip at the top shows the engine, the health of its data, what is standing and how the search is going; below it, a top five for each pace, with the evidence for each strategy.

How the search works

WHAT IT DOES	WHY IT MATTERS
Scores steadiness, not one good stretch	Half of a bot's training score is the *worse* half of the training years. A bot carried by one stretch scores low; a simple bot beats a complicated one that scores the same.
Scores AI rules honestly	A bot with a neural network, forest or other learned part is scored on training candles its model never saw, so a model that memorised its lessons no longer looks like an edge.
Keeps the best of every kind	The best bot of each kind — pace, type of strategy, long and short or long only — is kept. The search cannot collapse onto one idea, and when it gets stuck it restarts from these champions.
Learns which blocks work	It keeps count for every block — tried, faced the gauntlet, survived — and draws the ones that keep producing more often. **What is working** on the screen lists them.
Keeps one survivor of a kind	Variations of one idea — another stop, another look-back — are one idea. The first of a kind to survive holds its place; copies are not tested again. The work goes into new ideas, the lists stop filling with near-copies, and significance counts ideas, not copies.
Breeds from its survivors	Each generation, a few new bots are refinements of survivors, or two survivors merged into one.
Measures its own overfitting	The **overfit meter** asks: does the bot that trained best stay ahead on other data? Low is good. Near 50 % or more means the search is fitting noise — there may be no edge to find in those coins and paces.

Reading the top fives

EVIDENCE	WHAT IT MEANS
Proven on fresh data	15 or more trades in the vault and on the candles since, and in profit. The strongest evidence the edge finder gives.
Proving on fresh data	5 to 14 trades on fresh data, in profit so far.
Significant	Its result on the unseen data is strong enough to stand after every kind of bot tried (fluke odds under 10 %).
Candidate	It passed every check, but has too few trades on fresh data to judge and is not yet significant. Worth watching, not yet worth trusting.

The lists are worked out every 20 seconds, not at every bot, and a listed strategy keeps its place unless a newcomer is clearly better — so the screen holds still while thousands of bots are tested behind it. The strategy you are looking at stays on show until you click another.

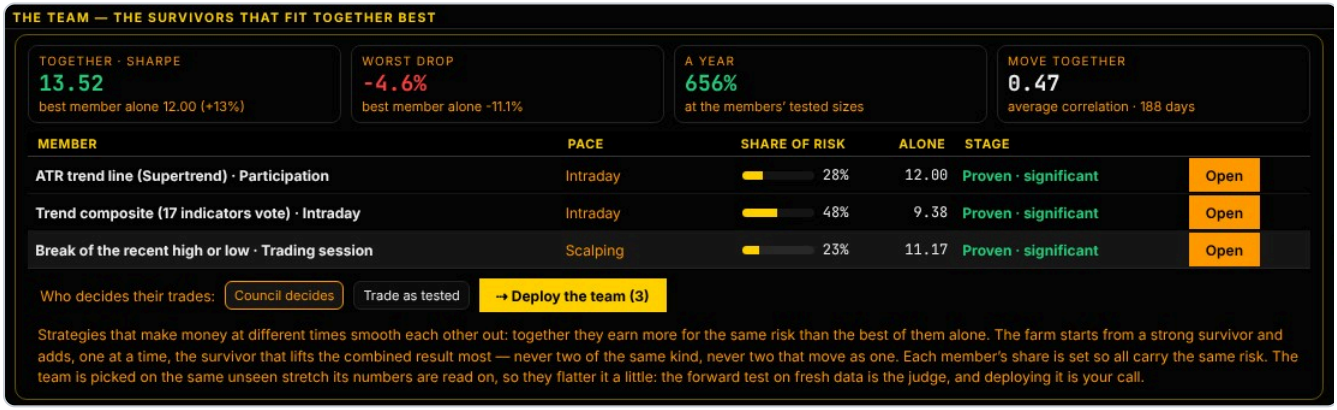
It looks after itself

The edge finder is built to run unattended for days.

- **If an exchange cannot be reached**, it tries the others. If none answers, it carries on with the candles it already has — it never drops a coin — and tries again after 1, 2, 5 and 10 minutes, then every 15 minutes, for as long as it takes. The **DATA** tile turns amber and says what it is doing. When the exchange is back, the fresh candles arrive and the tile clears. You do not need to stop and start it.
- **If the engine itself stops or hangs**, it is started again by itself from what was last saved — after 15 seconds, then a minute, then every five minutes.
- **Candles are refreshed** every half hour (every quarter of an hour for 15-minute paces), so the fresh-data tests keep moving.

The team: strategies that fit together

Strategies that make money at different times smooth each other out: together they earn more for the same risk than the best of them alone. The edge finder picks **the team** — up to five survivors, never two of the same kind, never two that move as one — and sets each member's share so all carry the same risk. It shows the team's Sharpe ratio, worst drop and a year's return beside the best member alone.



The team (here on test data): three survivors of different kinds and paces, their shares of the risk, and the combined result against the best member alone.

⇢ **Deploy the team** deploys every member to the crypto swarm after you confirm. The team is picked on the same stretch of data its numbers are read on, so they flatter it a little — fresh data is the judge.

Using it

- It starts by itself a minute and a half after the terminal opens, and carries on from where it was. Press ■ **Stop** when you are satisfied; it keeps everything it found and stays stopped until you press ▶ **Start**.
- ⚙ **Settings**: the coins, the paces to breed (scalping, intraday and swing by default; one-week, position and the 15-minute styles can be added), long and short or long only, the speed (*gentle* uses about half of one processor core and keeps the terminal smooth; *full* tries more bots an hour) and the drawdown limit for the shuffle test.
- Click a top-five strategy to see its rules in words, its curve, and every check it passed, with the numbers. Then **Open in the bot builder** to see what it is built of, block by block, or **Deploy to the crypto swarm** — choosing *Council decides* or *Trade as tested*. Deploying is always your decision.
- **The live fleet** shows every strategy you deployed, checked on the candles since: healthy, slipping or fading. When one fades, a top-five strategy at the same pace is offered as its replacement.

What it cannot promise

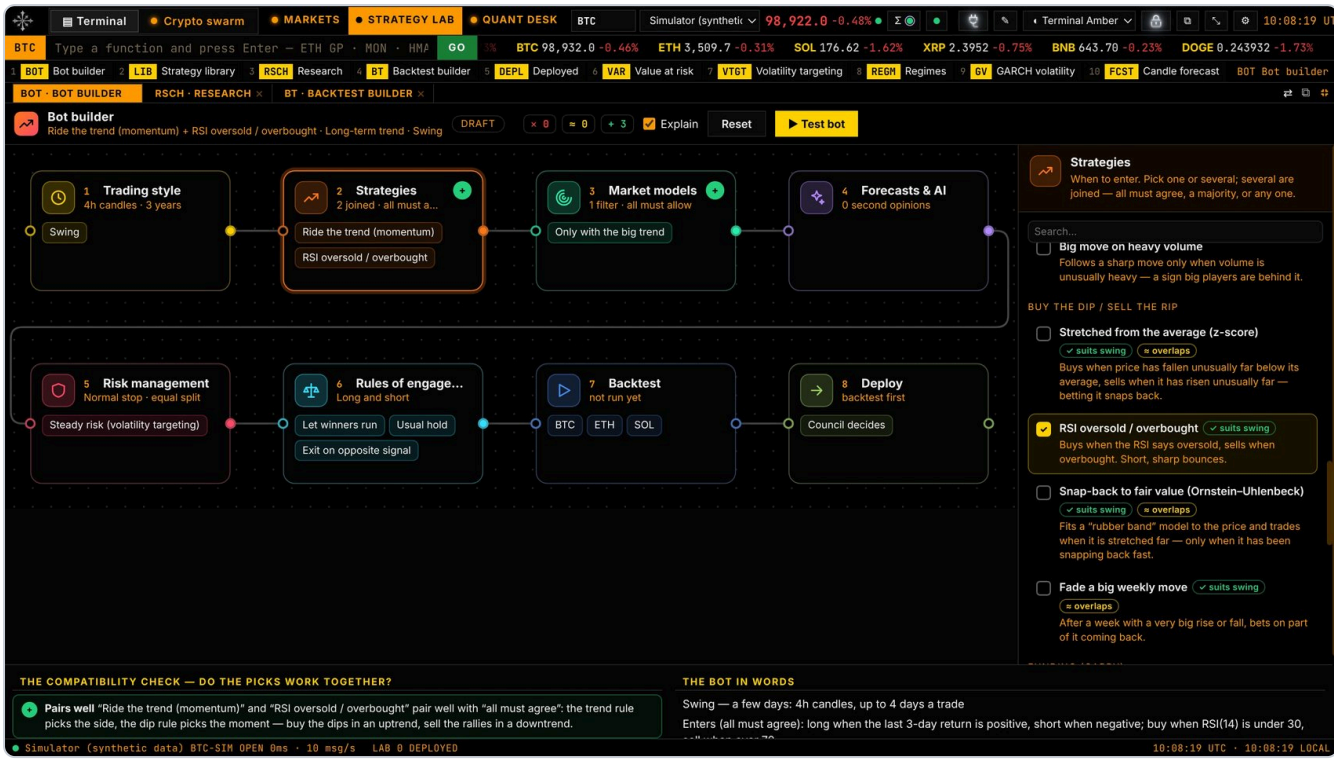
A bot that survived every check on history can still stop working, because markets change. That is why survivors keep being tested on fresh data, and why your deployed strategies are watched. Prefer strategies marked **proving** or **proven**; the checks lower the odds of a lucky bot, they do not make them zero. No software can promise profit: the edge finder's job is to throw out what does not hold up, and to show you honestly how strong the evidence is for what remains.

The Lab's screens

CODE	SCREEN	WHAT IT DOES
FARM	Edge finder	Breeds, tests and forward-tests bots by itself, all the time; lists the five best to deploy.
BOT	Bot builder	Build a bot block by block in plain English, backtest it and deploy it. The easiest way in.
RSCH	Research	Tests the whole library of published strategies on years of history, mixes them, and ranks the five best with an honest verdict.
LIB	Strategy library	Every rule and filter, the research papers behind it, and why it should work.
BT	Backtest builder	Build a strategy by hand from the library and test it.
DEPL	Deployed	The strategies trading in the crypto swarm: on or off, who decides their trades, and their record.

The bot builder, step by step

The bot builder is a canvas of eight connected blocks. Click a block and its options open in the panel on the right, each explained in plain English (tick **Explain**). A block can hold several choices.



The bot builder: eight blocks on the canvas, the selected block's options on the right, the compatibility check underneath.

1. **Trading style** — scalping or intraday on **15-minute candles**, scalping (hours) or intraday on 1-hour candles, swing (a few days), one-week trades, position (weeks to months) or long-term hold (up to a year). The style sets the candle size, look-backs, holding time, stop width and years of history for every other block.
2. **Strategies** — the entry rules, 36 in all: trend following, breakouts, buying dips and selling rallies, the day's own structure (VWAP, the opening range), timing and seasons, funding, trades against bitcoin, and AI (neural network, random forest, linear and next-candle forecasts). Pick one or several; several are joined in one of five ways (see "Five ways to join strategies" below). Where the neural network is picked, its size and how far ahead it looks can be set under it.
3. **Market models** — when the bot may trade: with the big trend, trending or ranging markets (ADX), statistical regimes (variance ratio, Hurst, hidden-Markov), calm or stormy volatility (GARCH), bitcoin's own trend (the market), activity, trading hours, weekdays or weekends, and funding crowding. Every filter you pick must allow a trade.
4. **Forecasts & AI** — second opinions that must agree, and an **AI decision maker** that learns from the bot's own past trades which signals to skip: the decision forest, the neural network, or both paired. It is kept only if it passes a check inside the training years (it must help on the last part of them). The unseen data is never used to choose it, so the result you read is not flattered by the choice.
5. **Risk management** — steady risk (volatility targeting), a daily loss cap (value at risk), half Kelly, storm-aware GARCH stops, a bitcoin hedge; two sizing rules can be combined (the smaller size wins). Also the stop width and how money is split across coins (equally, by calm, minimum variance or the efficient frontier).
6. **Rules of engagement** — long and short or one side only; let winners run or take profit at 1.5, 2 or 3 times the risk; holding time; whether an opposite signal closes the trade.
7. **Backtest** — choose the coins and press **Run backtest** (or ▶ **Test bot** at the top).

8. **Deploy** — as soon as a test is back, ⇢ **Deploy** lights up in the bot builder's header and beside the verdict (see "Deploying a strategy").

Strategies built on crypto research

Besides the classic rules, the library holds rules taken from published studies of crypto markets. Each shows its papers in `LIB`, and each must still pass the Lab's test on your coins.

STRATEGY	WHAT IT DOES
Momentum weighted by volume	Momentum where each candle counts by how much was traded; in crypto studies it beat plain momentum.
Trend composite (17 indicators vote)	RSI, stochastic, CCI, seven moving averages, MACD, volume trends, money flow and Bollinger bands each cast a vote. Based on the crypto trend factor published in 2025.
New 10-day high or low	Buys a close at a new 10-day high (trend) and at a new 10-day low (the bounce); holds a day; never shorts.
Follow a big day	After an unusually big daily move, trades in its direction for a day. The published edge is thin — test it.
Late-evening hours	Buys at 21:00 UTC and sells at 23:00 UTC, where bitcoin's gains were concentrated in 2015–2021. The pattern weakened afterwards.
Pairs trade: the coin against bitcoin	Buys a coin that is unusually cheap against bitcoin and sells one that is rich. With the bitcoin hedge (Risk block) it is a true market-neutral pairs trade.
Bitcoin leads, the coin catches up	When bitcoin has moved and a coin has not yet followed, trades the coin in bitcoin's direction.
Relative strength: coins beating bitcoin	Coins that have been beating bitcoin tend to keep beating it for weeks, and laggards keep lagging. Buys a coin that has outrun bitcoin by more than its usual wobble, sells one that has fallen behind — the opposite bet to the pairs trade.
Volatility squeeze breakout	Quiet markets do not stay quiet. When the bands around price are among the narrowest they have been in a long while, it goes with the first close outside them, until price falls back through its average.
ATR trend line (Supertrend)	A line trails price a few "average true ranges" away and only ever tightens. Long above it, short below; it sits through ordinary pullbacks and flips on an unusual reversal.

The three that read bitcoin's price (the pairs trade, bitcoin leads, relative strength) add BTC to the coins tested and never trade BTC itself.

Five ways to join strategies

JOIN	HOW THE STRATEGIES DECIDE TOGETHER
All must agree	A trade only when every strategy points the same way on the same candle. Fewer, better-confirmed trades.
Majority vote	More than half point the same way and none points the other way.
Any one	Any strategy fires, as long as none of the others disagrees. More trades.

Regime switch	The market's regime is read every candle (ADX): while it trends, only the trend and breakout strategies trade; while it ranges, only the dip-buying ones. Each strategy works in the market it was made for — the way to pair a trend rule with a dip rule.
Setup, then trigger	The first strategy sets the direction; the trade is taken when the others confirm it within the next few candles. A breakout and the volume that confirms it need not land on the same candle.

Rules with their own holding time — the two evening hours, one day for the big-day and new-high rules — are never held longer than that, whatever the trading style.

Intraday trading: 15-minute candles

Two styles trade on 15-minute candles: **Scalping — 15-minute candles** (trades over within a couple of hours, tested on six months of history) and **Intraday — 15-minute candles** (several trades a day, none held overnight, tested on a year). Every strategy and filter works at this speed, and four rules are made for it:

STRATEGY	WHAT IT DOES
VWAP trend	Long while price is above the day's volume-weighted average price (what the average buyer paid today), short while below.
Opening range breakout	The first two hours of the day set a range; trades the first close beyond it, in that direction, for the rest of the day. One trade a day.
Intraday momentum beyond the noise	Trades when the day's move is bigger than is normal for that time of day over the last two weeks.
Fade a stretched candle	After one unusually big candle on heavy volume, bets on part of it coming back within the hour.

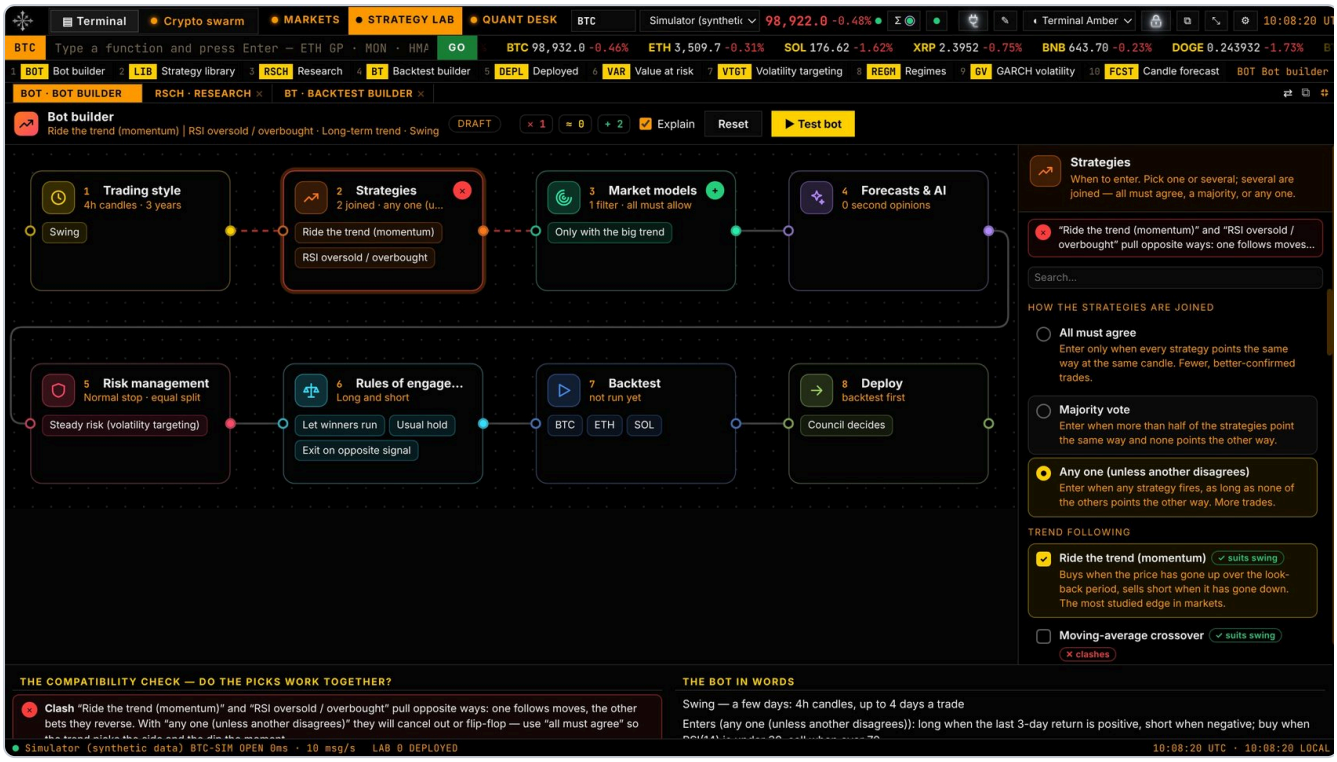
Faster is not automatically better

On 15-minute candles a bot trades many times a day, and every trade pays fees and slippage. Many ideas that look good on a chart make nothing after costs. The test counts them, and the line under "How its trades ended" tells you what share of the profit they took — if it says 80 % or more, the idea is too small for its speed.

The compatibility check

Every pick is checked against the others as you make it, and each option is tagged before you choose it:

- **✕ Clash** — the picks work against each other: for example a trend rule only allowed in a ranging market, or a breakout rule only allowed when it is quiet.
- **≈ Overlap** — they say the same thing: two trend rules, or a second opinion from the same model as the strategy.
- **+ Pairs well** — they add to each other: a trend rule with a dip rule when all must agree ("buy the dips in an uptrend"), or the neural network paired with the forest.



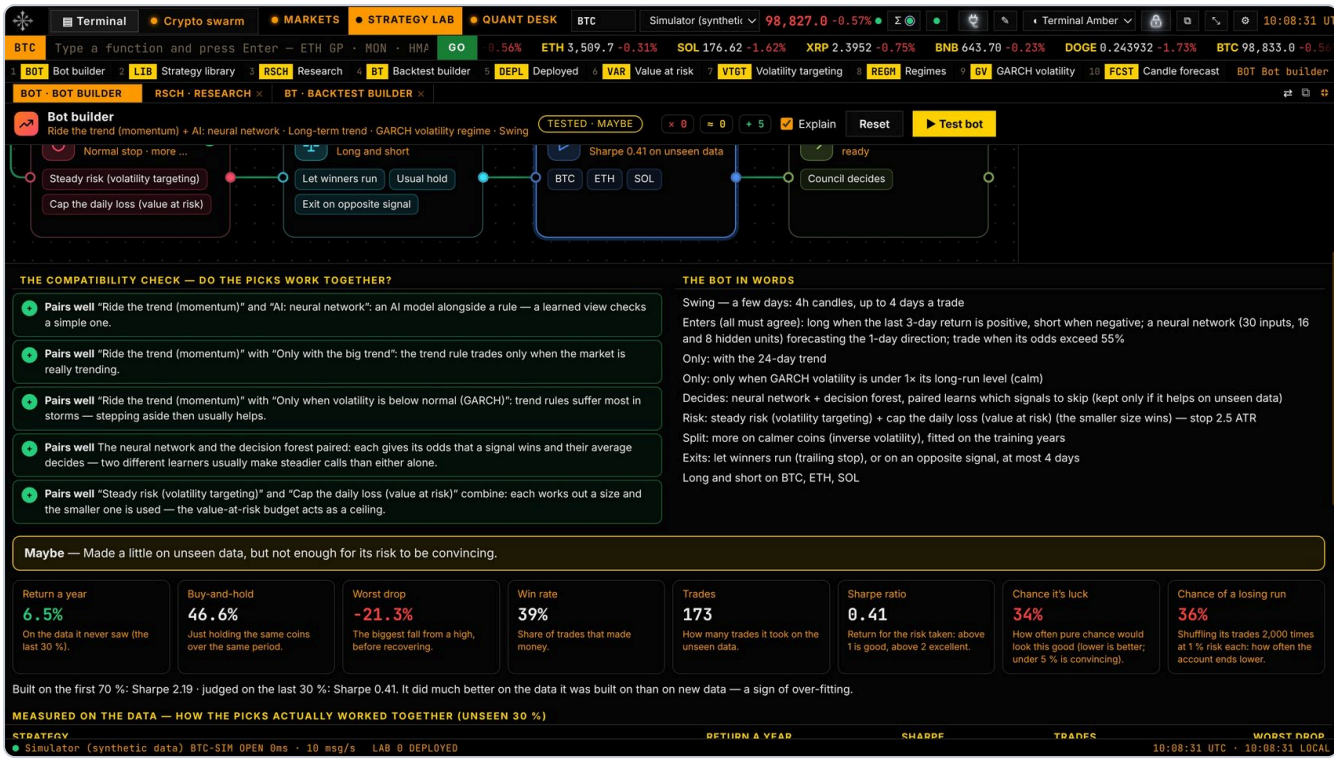
A clash flagged: a trend rule and a dip rule joined by "any one". The wires turn red and the check explains how to fix it.

Reading a backtest

Every bot is built on the first 70 % of the history and judged on the last 30 % it never saw, with fees and slippage. The result comes in plain words:

TILE	WHAT IT MEANS
Verdict	Looks good, Maybe, or Not yet — and why.
Return a year vs **Buy-and-hold**	The bot's yearly return on unseen data, against simply holding the same coins.
Worst drop	The biggest fall from a high before recovering (maximum drawdown).
Win rate and **Trades**	The share of winning trades and how many it took.
Sharpe ratio	Return for the risk taken: above 1 is good, above 2 excellent.
Chance it's luck	How often pure chance would look this good; under 5 % is convincing.
Chance of a losing run	Shuffling its trades 2,000 times: how often the account ends lower.

Below the tiles, **Measured on the data** shows each strategy alone and together, how two strategies actually related (correlation and how often they agreed), what each filter added, and how each AI decision maker did. If the bot did much better on the data it was built on than on new data, the Lab says so: that is a sign of over-fitting.

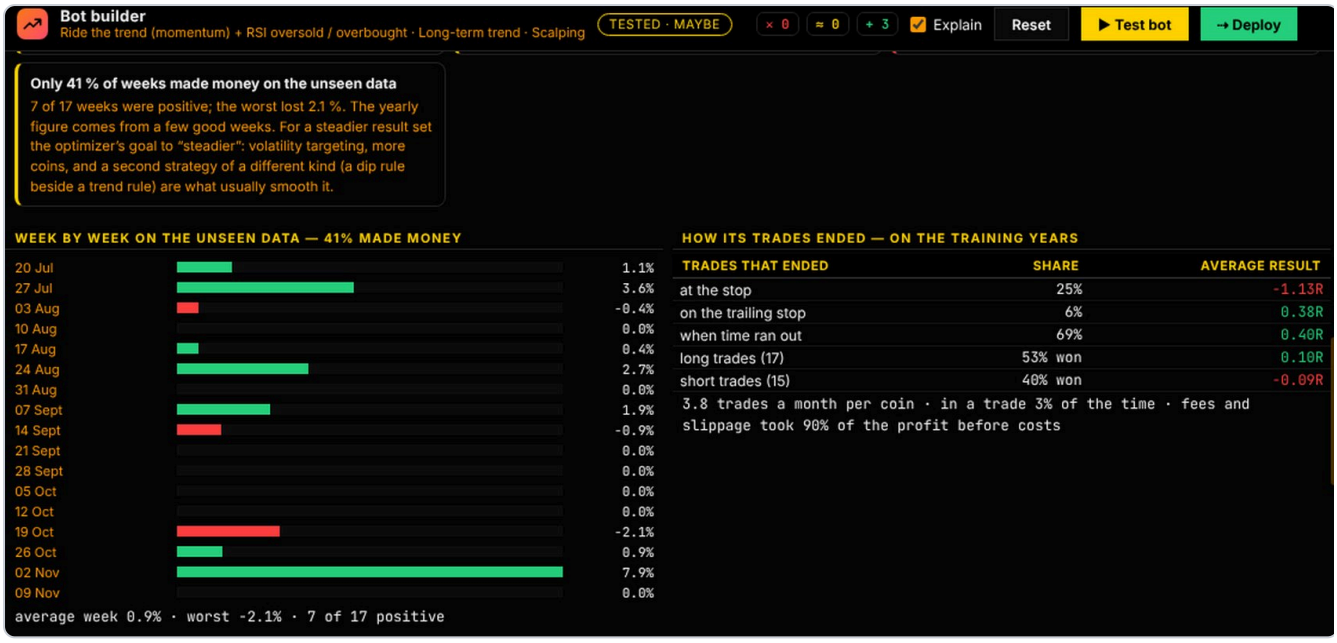


A tested bot: the compatibility check, the bot in words, the verdict and the result tiles.

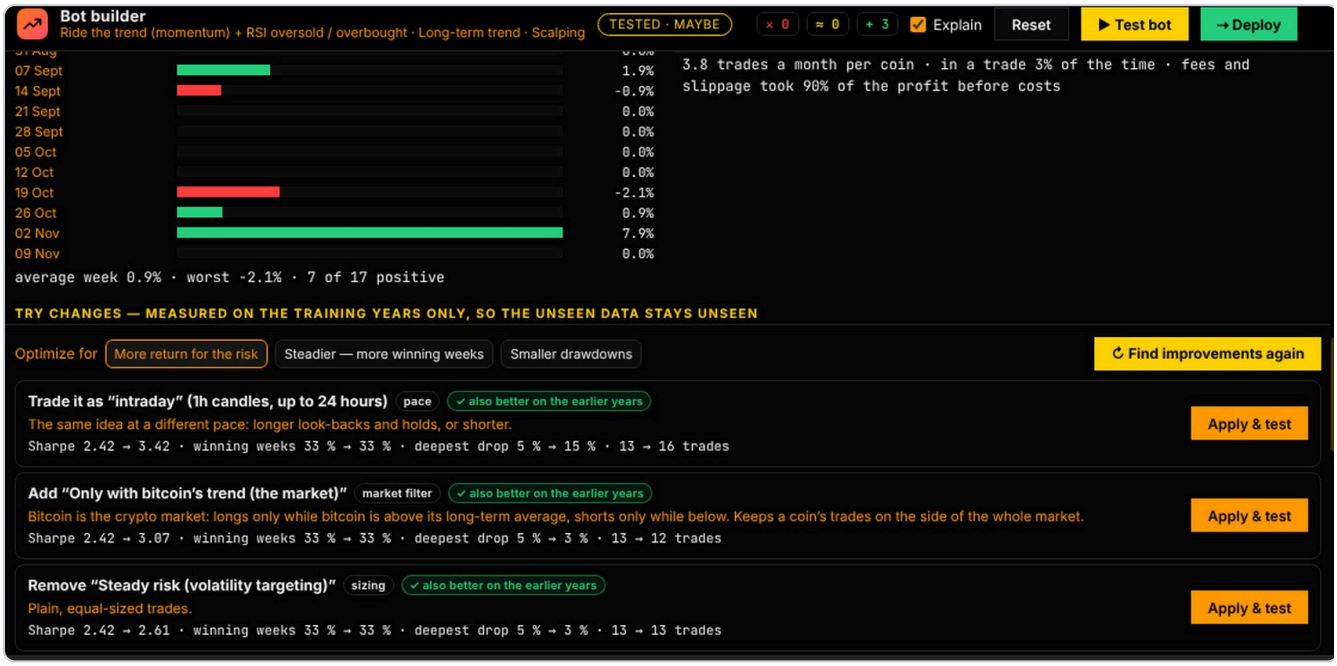
Optimizing a bot

Under every test result the Lab reads the result back to you and offers ways to improve the bot. It appears on its own as soon as a test finishes.

1. **Read what the test shows.** Cards in plain words: for example "70 % of trades end at the stop", "the short trades lose", "fees and slippage eat 90 % of the profit", "only 41 % of weeks made money". Each says what usually fixes it.
2. **Look at the result month by month** (week by week on short histories): each period's return, how many made money, and the worst one. This is what "a return every month" really looked like for this bot.
3. **Choose a goal**: more return for the risk, a steadier result (more winning weeks), or smaller drawdowns.
4. Press 🔍 **Find improvements**. The Lab makes every sensible one-step change to the bot — a wider or tighter stop, another exit, long only, each market filter added or removed, a second opinion, volatility sizing, each strategy's settings one step up and down, another pace — and measures each one.
5. Each change worth making is listed with its numbers before and after. Prefer the ones marked ✓ **also better on the earlier years**. Press **Apply & test** on one: the bot is changed and tested again.
6. When you are satisfied, press ⇢ **Deploy**.



Under a test: the diagnosis in words, the result week by week on the unseen data, and how the bot's trades ended on the training years.



The optimizer's list: each change, why it might help, the numbers before and after, and whether it also held on the earlier years.

Why it can be trusted

The optimizer never looks at the data your bot is judged on. It measures changes inside the training years only (fitting on the first part of them and scoring on the last part), so a change you apply still gets an honest test afterwards.

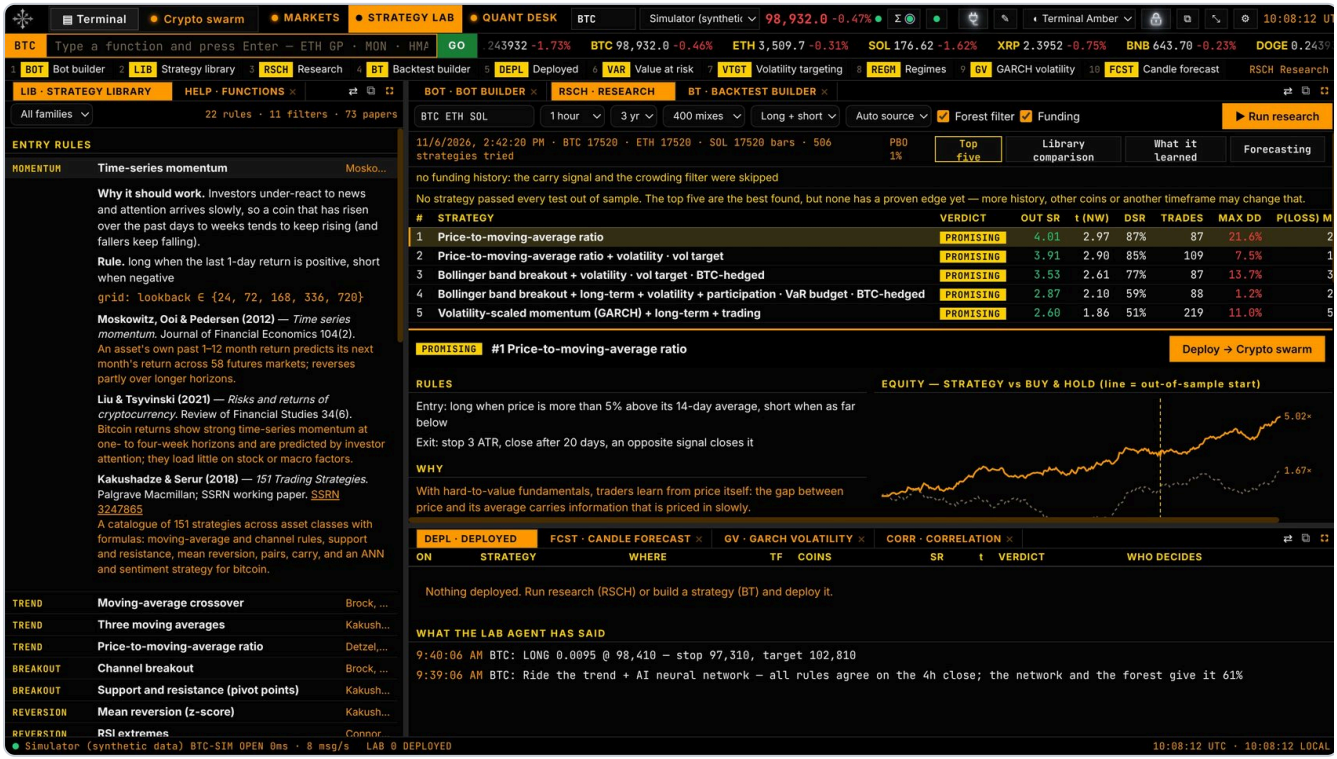
Do not over-optimize

Every version of a bot you test on the same unseen data makes it likelier that the best one looks good by luck. The Lab counts your versions and shows "chance the result is real after N tries" — watch that number, not just the Sharpe ratio. Apply one change at a time, and stop when the improvements get small. No optimizer can promise a profit every month; it can only show you how regular a bot has been and which changes made it steadier.

Research results have the same optimizer: open a ranked strategy in `RSCH`, press **Find improvements**, **Test this version**, then **Deploy this version**.

Research runs

1. Open `RSCH`. Choose the coins, the timeframe, the years of history and how many mixes to try.
2. Press **Run research**. The Lab downloads the history (cached for next time), tests every published rule, mixes the best with filters, exits and sizing, and sends the finalists to the unseen data.
3. Each finalist faces the out-of-sample Sharpe ratio, the deflated Sharpe ratio (corrected for how many strategies were tried), a random-entry test, a trade-shuffle test, a regression on buy-and-hold and simulated markets with no edge.
4. The top five get a verdict: **EDGE**, **PROMISING** or **NO EDGE** — said plainly when nothing passes.



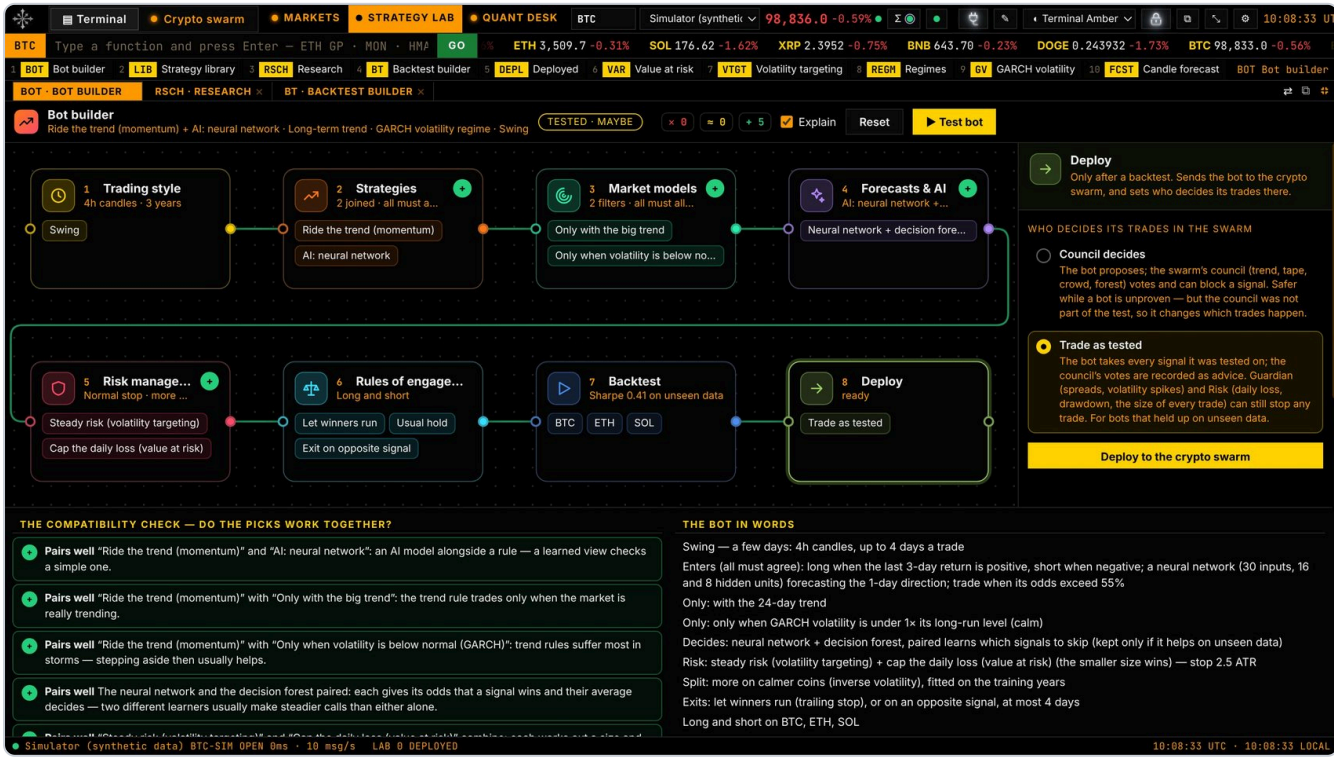
Research: the top five with their verdicts, the selected strategy's rules and its equity against buy-and-hold.

Deploying a strategy

1. Straight from the bot builder: once a test is back, press **⇢ Deploy** in its header or **⇢ Deploy to the crypto swarm** beside the verdict — the bot builder's test is the same test the backtester runs, so nothing else is needed first. You can also deploy from Research (**Deploy → Crypto swarm**) or from the swarm's deploy gate.

2. Choose **who decides its trades**:
   - **Council decides** — the strategy proposes; the swarm's council votes and the forest can veto. Safer while a strategy is unproven.
   - **Trade as tested** — the strategy takes the signals it was tested on; the council's votes are recorded as advice. For strategies that held up on unseen data.

   Either way, **Guardian** (spreads, volatility spikes) and **Risk** (daily loss, drawdown and the size of every trade) can always stop a trade.
3. In `DEPL` or the swarm's deploy gate: switch each strategy on or off, change who decides, or remove it. Each strategy keeps a record of its signals, trades, and any signals the council blocked — followed anyway, so you can see whether the council saved or cost money.



The Deploy block: choose who decides the bot's trades, then deploy to the crypto swarm.

> **Honest limits**
>
> A backtest, however careful, describes the past. Markets change and published edges fade. Out-of-sample results on one history are one draw: more coins, more years and another timeframe make a verdict firmer. Fills in fast markets can be worse than tested. Always run a new strategy on paper first.

CHAPTER 6

The crypto swarm

A team of agents that trades crypto perpetual futures on Hyperliquid — every decision debated, every trade sized and guarded.

How the swarm works

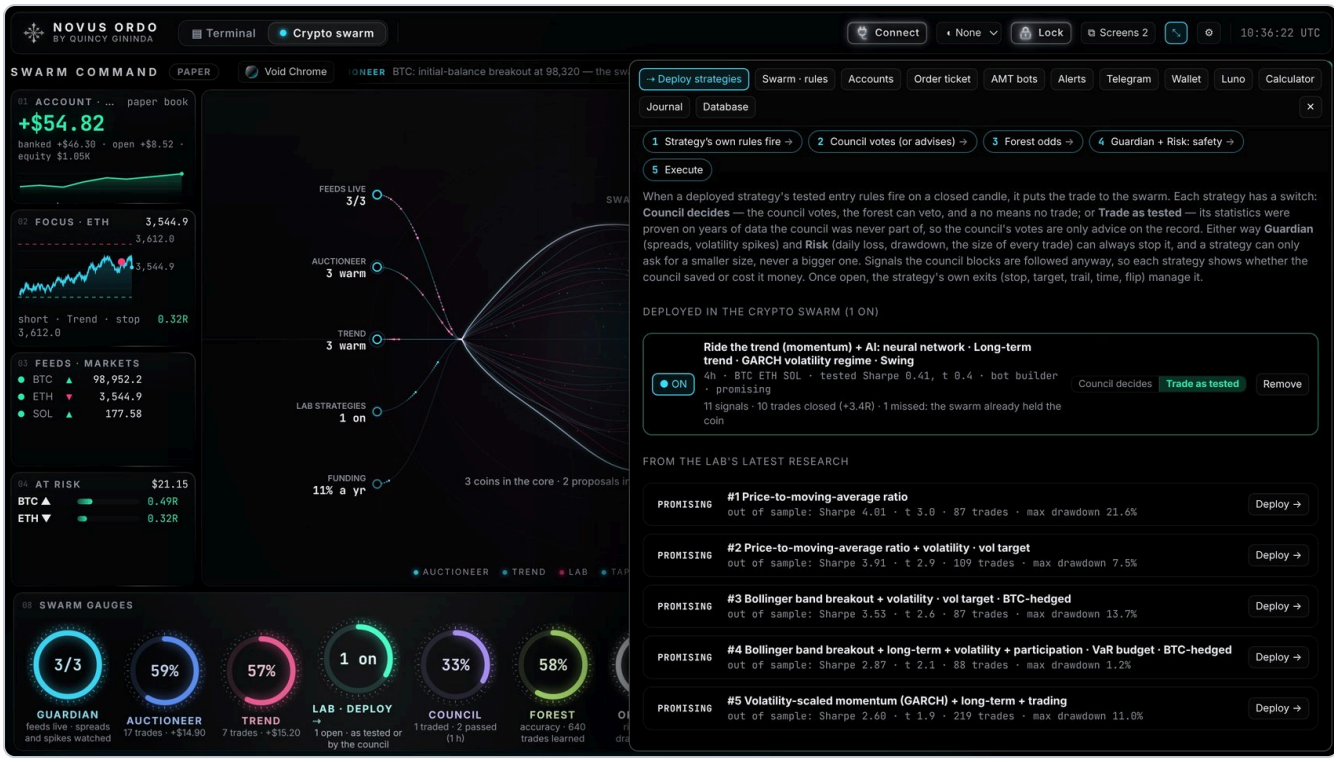
Three kinds of agent propose trades; five more vote on them. A trade happens only when no one vetoes it and the weighted vote is strong enough (unless a deployed strategy is set to trade as tested — then only Guardian and Risk can stop it).

AGENT	ROLE	WHAT IT DOES
Auctioneer	proposes	Intraday auction market theory: the session's value area, initial-balance breakouts, the 80 % rule back into value.
Trend	proposes	Time-series momentum: the 20-, 60- and 120-day trends vote; entries on hourly breakouts in their direction.
Lab	proposes	Your strategies deployed from the Strategy Lab.
Tape	votes	Aggressive buying and selling in the last seconds and the order book: for or against.
Crowd	votes	Funding: against a crowded side that pays a lot to hold, for when the other side pays.
Guardian	veto	Stops any trade when the spread is too wide or the market spikes.
Forest	votes, veto once trusted	A decision forest that learns from every trade (taken or refused) the odds of a loss, scratch, win or big win.
Risk	veto, sizes	The compounding risk manager: it decides the size of every trade and enforces the account's limits.

The Command Center

Open the swarm with **Crypto swarm** on the top bar or `Ctrl` + `2`.

- **Top** — the swarm's talk scrolling past, **⇢ Deploy**, **☰ Accounts & tools** and **Start / Stop swarm**.
- **Left** — the account and its profit curve; the coin in focus with its chart, entry, stop and target; the feeds; capital at risk.
- **Centre — the swarm core** — what the agents read flows in from the left; each coin is a sphere of particles that grows as its signals heat up; results flow out to the right. Click a coin to focus it.
- **Right** — equity, peak and drawdown; results; the execution tape of every fill.
- **Bottom — the gauges** — Guardian, Auctioneer, Trend, Lab · deploy, Council, Forest, Optimizer, Exposure, Loss limit and Floor; then the live feed of every proposal and vote.



The deploy gate: the decision chain, a deployed strategy set to "Trade as tested" with its record, and the Lab's latest research ready to deploy.

Running the swarm: paper, then live

1. **Paper first.** The swarm starts on a paper account trading Hyperliquid's real prices, with fees and slippage counted. Press **Start swarm** and watch the council work.
2. **Check the rules. Account · rules** (or `SWCF`): the coins, which agents may propose, risk per trade, leverage limits, daily loss limit, the floor.
3. **Deploy strategies** from the Strategy Lab (chapter 5) or the deploy gate.
4. **Go live.** Connect Hyperliquid (chapter 7), tick **Allow trading**, then in the rules choose **Live** and your key, and start the swarm. A dialog asks you to confirm. Every live position gets a stop on the exchange itself.
5. **Stop at any time. Stop swarm** stops new trades (open positions keep their exchange stops). The top bar's kill switch stops every bot; **Stop & close all** closes positions at market.

Risk control and compounding

- Every trade risks **1 % of what the account is worth now** (between 0.25 % and 2 % after adjustments): profits make the next trade bigger, losses smaller.
- Leverage is a consequence of the stop, capped at 3× per coin and 3× for all positions together, on isolated margin. One position per coin.
- After 20 closed trades, size follows the record (a quarter of the Kelly bet). Strong odds from the forest add size; weak odds take it off.
- Three-quarter size 10 % under the account's high; half size 20 % under.
- No new trades for the day after losing 5 %; a pause after three losses in a row.
- **The floor**: no trades under 70 % of the starting balance; once the account reaches 1.25× its start, the floor rises to 80 % of its high and only goes up — profits get locked in.

No system is always in profit

These rules keep losses small and survivable and let wins compound. They cannot remove market risk.

The swarm's tools

CODE	TOOL	WHAT IT IS FOR
TKT	Order ticket	Market, limit, stop and stop-limit orders on any connected exchange, with safety limits (chapter 8).
PORT	Portfolio	Balances, positions, open orders and fills on your connected accounts; cancel or close.
AMT	AMT bots	Auction-market-theory bots you run yourself, one per market: scalp, intraday, growth, order-flow hybrid, adaptive and daily trend styles, each with paper or live, backtest and training.
ALRT	Alerts	Price and order-flow alerts.
TG	Telegram	Your Telegram bot: trade cards and commands (chapter 7).
WALL	Wallet	Phantom wallet balances and activity (chapter 7).
LUNO	Luno lane	Crypto against South African rands on Luno: daily trend bots.
CALC	Calculator	Position size, risk/reward and liquidation price.
NOTE	Journal	Timestamped trading notes.
DB	Database	Recorded trades, retention and CSV export.

Trade reports on your phone

With Telegram connected, every swarm trade sends three messages, each with a chart: **the setup** (entry, stop, target, the potential against the risk, the odds and the council's votes), **managing** (the first decision after entry, such as moving the stop to break-even) and **the result** (profit or loss in dollars and R, how long it was held, fees, and a verdict in plain words).

CHAPTER 7

Wallets and connections

Connect exchanges, wallets and Telegram — safely. Everything starts from 🔌 Connect on the top bar.

Connection options

CONNECTION	WHAT IT IS USED FOR
Hyperliquid	Crypto perpetual futures — the crypto swarm's exchange, the order ticket and bots. Your account is a wallet; its money is USDC. No documents, no IP whitelist.
Luno (ZAR)	Buying and selling crypto with South African rands; the Luno lane's daily trend bots.
Binance, Binance Futures, Coinbase, Bybit, OKX, Kraken, Bitget, KuCoin	Balances, positions and orders through the portfolio and the order ticket; AMT bots.
Any CCXT exchange	More than 100 further exchanges, through the same vault.
Telegram bot	Trade cards and commands on your phone.
Phantom wallet	Read-only view of a Solana wallet's balances and activity (`WALL`).
Solana RPC (optional)	A private Solana connection for heavy wallet use; the app works without it.

How keys are kept safe

- Keys are **encrypted with your operating system's key store** (Windows DPAPI) before they reach the local database. The screen only ever shows masked values.
- Keys are used only by the desktop process; they never leave your computer except to sign requests to that exchange.
- Every key is **read-only until you tick "Allow trading"** on it. A key marked **testnet / sandbox** only trades the exchange's testnet.
- **Never enable withdrawals or transfers** on any exchange key. The terminal never needs them.
- For Hyperliquid, the terminal uses an **API wallet** key, which can trade but can never withdraw. It refuses your main wallet's own key.

Hyperliquid: funding and connecting

Three rules that protect your money

1. **Never** send USDC from an exchange straight to Hyperliquid: it credits the wallet that sent it — the exchange's — and the money is lost. Always go through your own wallet.
2. **Never** give anyone, or any app, your wallet's recovery phrase or main private key.
3. Deposit **at least 5 USDC** each time; smaller deposits are not credited.
4. **Never** send a Hyperliquid withdrawal straight to Luno: Hyperliquid pays out USDC on Arbitrum, and Luno lists USDC deposits on Ethereum only.

There are two ways to get USDC into Hyperliquid. Both start with a wallet of your own (step 1), which is also your Hyperliquid account.

ROUTE	HOW	COST
A · Card, inside Hyperliquid (quickest)	`app.hyperliquid.xyz` → **Connect** → **Deposit** → **Fiat**; pay by card or bank transfer through the payment company (Swapped), which may verify your identity the first time. Check the screen says it lands in your **Hyperliquid balance**.	No Hyperliquid fee; about 2 % in the price, often 3–5 % in all by card, plus any foreign transaction fee from your bank. The total shows before you confirm.
B · Through an exchange (cheapest)	Buy USDC on Binance → withdraw to **your own wallet** on **Arbitrum One** → **Deposit** on Hyperliquid.	The exchange's trading fee and cents of ETH for network fees.

1. **Get a wallet of your own** — the **Rabby** or **MetaMask** browser extension on the computer that runs Novus Ordo is the most reliable. Keep the recovery phrase on paper.
2. **Fund it — route A**: `app.hyperliquid.xyz` in the same browser → **Connect** → **Deposit** → **Fiat**, pay by card, start small. **Or route B**: buy USDC (plus about $2 of ETH) on your exchange, withdraw both to your own wallet on **Arbitrum One** with a small test first, then **Connect** → **Deposit** on Hyperliquid (5 USDC or more).
3. **Keep a little ETH on Arbitrum** in your wallet (about $1–2) — needed to send USDC on to Binance when you cash out. Buy it with the wallet's own **Buy** button or withdraw it from Binance on Arbitrum One.
4. **Create the API wallet**: **More** → **API** → name it Novus Ordo → **Generate** → **Authorize API Wallet**. Copy the private key: it is shown once.
5. **Connect it**: 🔌 **Connect** → **Hyperliquid**. **Main wallet address** = your wallet's 0x... address; **API wallet private key** = the key you copied. Save, then **Test live**.
6. **Allow trading** when you are ready, then choose **Live** in the swarm's rules.
7. **Withdrawing** is done only on `app.hyperliquid.xyz` with your own wallet — the terminal cannot withdraw.

Taking money out — back to rands

1. On `app.hyperliquid.xyz` : **Withdraw** → amount → sign. The USDC arrives in your wallet on Arbitrum a few minutes later (Hyperliquid charges a small flat fee).
2. On Binance: **Deposit** → **USDC** → **network Arbitrum One**, copy the address, and send the USDC to it from your wallet (a test amount first).
3. Sell the USDC for rands on Binance and withdraw to your bank.

Not to Luno — unless it shows Arbitrum

Luno's help pages list USDC deposits on the Ethereum network only, so USDC sent to Luno on Arbitrum will most likely not arrive. Before ever using Luno for this, open the Luno app → **Receive** → **USDC** and use it only if **Arbitrum** is listed. Hyperliquid's own cash-out to a bank is reported to be coming; when it supports rands, this gets simpler.

Practise first

Make an API wallet on `app.hyperliquid-testnet.xyz` (its faucet gives free test USDC), tick **testnet / sandbox** on the key, and run the swarm live against the testnet.

Luno, Telegram and Phantom

Luno (South African rands)

1. On luno.com (the website): **Settings → Security → API keys → Create API key**.
2. Choose the permission that can view balances and place and cancel orders — **never** one that can send, withdraw or transfer.
3. Paste the Key ID and Secret into 🔌 **Connect → Luno**, press **Test live**, and tick **Allow trading** only when you want orders placed.

Telegram

1. In Telegram, open **@BotFather**, send `/newbot`, choose a name and a username ending in "bot".
2. Paste the token into 🔌 **Connect → Telegram bot**.
3. Open the Telegram panel (`TG`), press Start in your bot's chat, then **Detect chat** and **Send test message**.
4. The bot also answers `/price`, `/status` and `/help` — only from your own chat, while the terminal is open.

Phantom wallet

Connect Phantom in the Wallet panel opens a one-time page in your browser, where you approve the connection and sign a message. The terminal verifies the signature and then shows the wallet's SOL and token balances, their value and recent activity. It is read-only: the terminal never sees a private key and cannot move funds.

CHAPTER 8

Placing orders and managing accounts

The order ticket, the portfolio and the safety rules that protect every order.

The order ticket

1. Open `TKT`. Choose the connected key (exchange account) and the market.
2. Choose the order type: **Market**, **Limit**, **Stop** or **Stop limit**. Size it in the coin or in dollars, or use 10/25/50/100 % of your free balance.
3. Options: post-only, reduce-only, time in force (GTC, IOC, FOK) and leverage on derivatives. Clicking a level on the `DOM` sends its price to the ticket.
4. Review the confirmation, which shows the full order with a **LIVE** or **TESTNET** badge, and send.

Built-in order safety

These rules run in the desktop process, not the screen, so nothing can bypass them:

RULE	DEFAULT
A key places orders only after **Allow trading** is ticked on it	off until you tick it (cancelling always works)
Maximum value of one order	$1,000 (reduce-only and close orders are exempt: they can only shrink a position)
Rate limit	10 orders a minute
Maximum distance from the market price	50 %
Confirmation before sending	on (can be switched off under ⚙ in the ticket)
Testnet keys	never fall back to live
Order log	every attempt — placed, rejected or failed — is recorded

Stops that would trigger immediately, post-only on non-limit orders and reduce-only on spot are refused before they reach the exchange.

Portfolio, calculator, alerts and journal

- `PORT` — balances, positions, open orders and fills; **Cancel**, **Cancel all** and **Close** (reduce-only at market).
- `CALC` — position size from risk, reward-to-risk, and liquidation price before you trade.
- `ALRT` — price and order-flow alerts that pop up on screen.
- `NOTE` — a timestamped trading journal.

CHAPTER 9

Security and safety

How Novus Ordo protects your keys, your account and your data.

Security at a glance

PROTECTION	HOW IT WORKS
Ledger key lock	An 8-digit key, stored only as a salted digest. Until it is entered, the desktop process refuses keys, trading, bots, wallets and Telegram. Wrong attempts lock input for longer each time.
Encrypted key vault	Exchange keys are encrypted by the operating system's key store before they are stored; the screen shows only masked values; copying a key clears the clipboard after 30 seconds.
Read-only by default	No key can place an order until you tick Allow trading on it.
No withdrawals	The terminal never needs, and never uses, withdrawal permission. Hyperliquid uses an API wallet that cannot withdraw.
Order limits	Order value, rate and distance limits, confirmations and a full order log (chapter 8).
Risk limits	The swarm's daily loss limit, drawdown cuts and rising floor; exchange-side stops on live positions (chapter 6).
Kill switch	One click on the top bar stops every bot.
Local data	Your database, settings and keys live on your computer only.

Good practice

- Use a strong, private ledger key, and press 🔒 **Lock** when you leave the computer.
- Create exchange keys with trading permission only; restrict them to your IP address where the exchange offers it.
- Keep wallet recovery phrases on paper, offline, and never type them into any application.
- Start on paper, then testnet, then live with a small amount; raise size only after a record you trust.
- If you think a key has leaked, delete it on the exchange and in 🔌 Connect, and create a new one.

CHAPTER 10

Troubleshooting

Quick answers to the most common problems.

Common problems

PROBLEM	WHAT TO DO
"Novus Ordo is already open" or "database is locked"	Another copy is still running or closing. Press `Ctrl`+`Shift`+`Esc`, end every **Novus Ordo** task in Task Manager, then press **Try again** or open it again. Your data is safe. Use only one copy (zip or portable, not both).
The window stays blank or black	If it is a graphics-driver problem, the terminal restarts itself in **safe mode** (no GPU). You can also choose **Help → Restart in safe mode**. If it persists, send the log (below).
The portable version starts slowly	It unpacks itself on every launch while Windows Defender scans it. Use the zip version for fast starts.
A feed shows "error" or no data	Check your internet connection; some exchanges are restricted in some countries — choose another venue. The terminal reconnects by itself.
An exchange rejects a key	Use **Test live** in 🔌 Connect: it explains the exchange's error in plain words (wrong permission, IP restriction, clock difference). Never send anyone your key — send the error message.
A key shows "(read-only)"	Tick **Allow trading** on it in 🔌 Connect.
Hyperliquid won't connect to a phone wallet	Use the Rabby or MetaMask browser extension on the computer instead (chapter 7), or open the site inside the wallet app's own browser.
USDC sent from Hyperliquid has not arrived	Check the network: Hyperliquid pays out on Arbitrum, so the receiving address must be an **Arbitrum** USDC address (Binance: Arbitrum One). If it went to an address on another network, contact that exchange's support with the transaction link.
Windows SmartScreen warning	**More info → Run anyway**.

The log file

Help → Open log folder shows `novus.log`, which records start-up timings, any delay of the app and any error. When reporting a problem, send this file — never your keys.

Location on Windows: `C:\Users\<you>\AppData\Roaming\novus-ordo\logs\novus.log`

CHAPTER 11

Risk disclosure and terms of use

Please read before trading with real money.

Risk disclosure

Cryptocurrencies are highly volatile. Perpetual futures use leverage: a small move against a position can cause a large loss, and a position can be liquidated, losing all the margin committed to it. Past performance — in a backtest, a paper account or live trading — is not a reliable indicator of future results. Statistical tests and forecasts estimate probabilities; they do not remove risk.

Automated trading carries additional risks: software errors, connection failures, exchange outages, delays and fills worse than expected in fast markets. Monitor live trading, keep limits conservative, and use the kill switch when in doubt.

Not financial advice

Novus Ordo is software that displays market data, runs calculations and executes instructions you configure. It is not an investment adviser, broker, exchange or custodian, and nothing in the terminal or this manual is a recommendation to buy or sell any asset. You are solely responsible for your trading decisions, for the keys and permissions you grant, and for complying with the laws and tax obligations where you live. Exchange availability varies by country; use only services that are permitted for you.

Your responsibilities

- Keep your ledger key, exchange keys and wallet recovery phrases secret.
- Never grant withdrawal permission to any key used with the terminal.
- Test on paper and testnet before trading live; trade only money you can afford to lose.

APPENDIX A

Function code reference

Every function, by code. Type the code on the command line; put a coin first to switch market (`ETH GP`).

All functions

CODE	FUNCTION
MON	Monitor
HMAP	Heat map
GP	Graph price
GIP	Footprint
LHM	Liquidity heat map
BOOK	Order book
DOM	Depth ladder
DPTH	Depth chart
TAS	Time & sales
CVD	Cumulative delta
FLOW	Flow pulse
BLCK	Block trades
LIQ	Liquidations
VP	Volume profile
OI	Open interest
FUND	Funding
OMON	Options
SPRD	Venue spread
CORR	Correlation
HVOL	Volatility
GV	GARCH volatility
FCST	Candle forecast
DES	Description
SCRN	Screener
WL	Watchlist
TOP	Top news
WEI	World markets

CODE	FUNCTION
FARM	Edge finder
BOT	Bot builder
RSCH	Research
LIB	Strategy library
BT	Backtest builder
DEPL	Deployed strategies
VAR	Value at risk
CAPM	CAPM
EF	Efficient frontier
KELLY	Kelly criterion
NNET	Neural network
PAIR	Pairs & cointegration
SEAS	Seasonality
MC	Monte Carlo
DHG	Delta hedging
VTGT	Volatility targeting
MOM	Momentum
REGM	Regimes
SWRM	Crypto swarm view
SWCF	Swarm rules
TKT	Order ticket
PORT	Portfolio
AMT	AMT bots
ALRT	Alerts
TG	Telegram
WALL	Wallet
LUNO	Luno lane
CALC	Calculator
NOTE	Journal
DB	Database
HELP	Every function

APPENDIX B

Glossary

The terms used in the terminal, in plain English.

Terms A to Z

ATR (average true range)
The average size of a candle's range; used to set stops that fit how much a coin normally moves.

Backtest
Running a strategy on past data to see how it would have done.

Beta
How much a coin tends to move when the market moves; beta 1.5 moves 1.5 % for the market's 1 %.

Cointegration
Two prices are cointegrated when a mix of them keeps returning to its average, however far each wanders alone — the basis of a pairs trade.

CVD (cumulative volume delta)
Running total of buying minus selling volume.

Delta
In order flow, buying minus selling volume; in options, how much the option's price moves with the coin.

Drawdown
A fall from the account's high before it recovers.

Efficient frontier
The mixes of assets with the best expected return for each level of risk.

Expected shortfall
The average loss on the days worse than the value at risk.

Footprint
A chart showing the volume bought and sold at each price inside each candle.

Funding
Regular payments between long and short holders of a perpetual future, keeping its price near the spot price.

GARCH / GJR-GARCH
Models that forecast volatility from how it clusters; GJR adds the larger reaction to falls.

In-sample / out-of-sample
Data a strategy was built on, and data it never saw and is judged on.

Isolated margin
Each position has its own margin, so it can only lose that margin.

Kelly criterion
The bet size that grows capital fastest in the long run; half Kelly is the common, safer choice.

Leverage
Trading a position larger than the money put up; it magnifies gains and losses.

Liquidation
The forced closing of a leveraged position when its margin runs out.

Monte Carlo
Simulating thousands of possible futures from a strategy's trades to see the range of outcomes, not just the average.

Neural network
A machine-learning model of connected artificial neurons that learns patterns from data.

Open interest
The number of open futures contracts.

Order flow
The actual buying and selling — who is aggressive, at what prices, in what size.

Over-fitting
A strategy tuned so closely to past data that it fails on new data.

Pairs trade
Buying one asset and selling a related one, betting the gap between them closes; the market's direction cancels out.

Opening range
The high and low of the first hour or two of the trading day; a break of it often sets the day's direction.

Paper trading
Trading with simulated money at real prices.

Perpetual future
A futures contract with no expiry, kept near the spot price by funding.

POC (point of control)
The price with the most volume traded in a session.

R (risk unit)
The amount risked on a trade; a 2R win made twice what was risked.

Random forest / decision forest
A machine-learning model made of many decision trees that vote.

Seasonality
A pattern tied to the clock or calendar: an hour of the day or a day of the week that tends to rise or fall.

Sharpe ratio
Return per unit of risk; above 1 is good, above 2 excellent.

Slippage
The difference between the expected price and the price actually filled.

Stop (stop-loss)
An order that closes a trade if price moves against it by a set amount.

Testnet
An exchange's practice environment with free test money.

Value area
The price range where 70 % of a session's volume traded.

Value at risk (VaR)
The loss not expected to be exceeded at a chosen confidence over a chosen time.

VWAP
Volume-weighted average price: the average price paid today, counting each trade by its size.

Volatility
How much price moves; usually given as a yearly percentage.

Volatility targeting
Sizing positions so the portfolio's risk stays near a chosen level.

CREDITS

NOVUS ORDO

The professional crypto trading terminal

CONCEIVED, DESIGNED AND CONSTRUCTED BY

Quincy Gininda

The market terminal, the Quant desk, the Strategy Lab and bot builder, the crypto swarm and its agents, and this manual are the work of Quincy Gininda.

Operator's Manual · Edition 1.6 · November 2026
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